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POLICY WATCH

Economic geography, globalisation and New Zealand's productivity paradox

Philip McCann*

*Department of Economics, University of Waikato, Private Bag 3105, Hamilton 3240,
New Zealand*

This paper examines New Zealand's poor productivity performance from the reform period onwards, from the perspective of economic geography. Rather than employing institutional or free-market versus interventionist arguments to explain New Zealand's low productivity, as is usually the case, the argument developed here is that the debate should be considered from a very different viewpoint. If we adopt an economic geography perspective, there is nothing really paradoxical about New Zealand's productivity performance. As such, New Zealand's productivity performance is rather more of a conundrum, a riddle, with a fairly straightforward solution.

Keywords: productivity; cities; trade; migration; geography

1. Introduction

The economic performance of New Zealand in many respects appears to be something of a paradox. Simply put, in spite of wide-ranging and far-reaching competitive reforms of the 1980s and 1990s, the productivity performance of New Zealand has remained stubbornly mediocre in comparison to many other advanced economies. As the 2003 OECD economic survey of New Zealand noted '... The mystery is why a country that seems close to best practice in most of the policies that are regarded as the key drivers of growth is nevertheless just an average performer' (OECD, 2003). This mediocre performance leads to spirited debates between commentators in the press, academic and professional researchers, and public policy-makers, as to the desired policies that the country should adopt. Within these debates there are broadly two camps. One camp focuses on institutional issues and tends to encourage further deregulation and liberalisation, while another camp sees greater public regulation and public intervention and labour market protection as the way forward. For the purposes of this paper, the former camp can be labelled the 'greater-deregulation' camp, while the latter camp can be labelled the 'greater-intervention' camp. Interestingly, in both cases, each of the two opposing camps draws their analyses, inferences and conclusions to a large extent on a different reading of New Zealand's economic history.

*Email: pmccann@waikato.ac.nz

The argument to be put forward in this paper is that neither the greater-deregulation nor the greater-intervention approaches provides even approximately a suitable diagnosis of the New Zealand's current situation and predicament, and, without a suitable diagnosis, an appropriate remedy cannot be developed. In contrast to the two currently-opposing approaches, we here develop an explanation of New Zealand's current economic situation, which is based on the arguments of economic geography and, in particular, on the economic history of economic geography, as specifically applied to the context of New Zealand. It will be argued here that a coherent understanding of New Zealand's current predicament can be gained by examining the modern interrelationships between geographical location, economies of scale, and the diversity of production and trade. These interrelationships are the key themes in modern economic geography, and largely underpin Paul Krugman's recent Nobel Prize in Economics. To be fair, these are topics that the New Zealand economics profession as whole, and in particular economists at The New Zealand Treasury and Reserve Bank of New Zealand, have traditionally had very little expertise or interest in. Yet, it will be argued here that economic geography can provide a clear explanation of many of the apparently paradoxical features of New Zealand's performance. This suggests that researchers seeking to explain New Zealand's relative productivity performance need to take a much fuller and more systematic account of geographical factors than they are accustomed to doing. New Zealand's productivity performance will be seen to be less of a paradox and rather more of a conundrum, a particular riddle, with a very particular set of analytical explanations. These explanations also point to particular possible policy solutions.

This paper is organised as follows. The next section examines New Zealand's apparent productivity paradox and the various arguments as to why New Zealand's productivity performance has been less than expected. Section 3 discusses the major features of the current phase of globalisation and Section 4 discusses some of the major insights from modern economic geography. Section 5 then applies the arguments of Sections 3 and 4 to the case of New Zealand. This allows us to demonstrate that New Zealand does not really exhibit a productivity paradox. Rather, according to economic geography theory, the labour productivity performance of New Zealand is more or less exactly as expected, as are a range of other currently-observed characteristics of the New Zealand economy. This assessment therefore provides for a range of potential policy directions as discussed in Section 6.

2. New Zealand's productivity paradox

The standard of living of a country can obviously be gauged in different ways, including measures of income distribution, life-expectancy, crime statistics, health indices, and measures of wellbeing and happiness (Layard, 2005). However, the most commonly used index is output per capita, represented by either GDP per capita or income per capita, which itself is used as an approximation of output per worker. This is generally regarded by economists as being the best comparison measure of wealth and living standards between countries. As Krugman (1997) remarks, 'Productivity isn't everything, but in the long run, it is almost everything. A country's ability to improve its standard of living over time depends almost entirely on its ability to raise its output per worker'. For comparisons between countries and across time various different measures of GDP per capita are used in different

contexts, including Geary-Khamis dollars (Maddison, 2006), the Atlas method,¹ the Laspeyres Index (Penn World Tables 6.2), GDP per capita purchasing power equivalents (OECD, 2008a), current US dollar income per capita purchasing power parity equivalents (OECD, 2008a; World Bank, 2008a). However, whatever method of output or income per capita is employed, the history of New Zealand's labour productivity performance is fairly clear. During the three decades following the Treaty of Waitangi, New Zealand was the fastest growing economy in the world (Maddison, 2006). Indeed, by the year 1914, on the eve of the First World War, New Zealand's GDP per capita was the highest in the world (Maddison, 2006). During the first four decades of the 20th century, the UK maintained a sophisticated and highly integrated set of bilateral trade, capital mobility and labour mobility relations with each of its dominions. The GDP per capita levels of the UK, New Zealand, Canada and Australia all stayed very close to one another (Maddison, 2006), as would be expected from a regime in which both labour and capital were highly mobile between countries (Borts & Stein, 1964). These relative parity relationships, however, were significantly affected by the Second World War as the infrastructure capacity and performance of the UK economy suffered immensely during this conflict. As with many of the war-ravaged European economies, it took almost 15 years for the UK economy to begin to function again normally (Eichengreen, 2007), in part due to help from the Marshal Plan (Findlay & O'Rourke, 2007). In contrast, an absence of significant infrastructure damage allied with increased export output war-demands during the Second World War and the Korean War (Gould, 1982) meant that, even as recently as 1960, New Zealand was still the world's third highest GDP per capita country (Maddison, 2006), behind only the USA and Switzerland. As well as the explicit bilateral migration-promotion policies developed between the UK and its dominions, the post-war productivity gap between the UK and its dominions also largely accounted for the scale and success of the mass-migrations from the UK to its dominions during the 30 years after the war.

During the post-war reconstruction era the productivity growth performance of many of the other OECD countries superseded that of New Zealand (Gould, 1982), the result of which was that the gap in productivity levels between New Zealand and other OECD countries rapidly closed. In part this was a result of the fall in the terms of trade of New Zealand's primary exports (Easton, 2007). It was also in part due to the fledgling processes of European integration, which rapidly boosted the productivity performance of many European OECD countries and accelerated the UK's entry into the EEC (Eichengreen, 2007). In response to these trends, New Zealand had been moving slowly towards a more liberal economic and monetary regime right up until the second oil price shock of 1981 (Singleton, 2006), and export diversification had increased markedly (Easton, 1994). The hesitant market liberalization process, however, became somewhat stalled during the early 1980s (Easton, 1994). Finally, in 1984, New Zealand's experiment with free market reforms began in earnest. The reforms took place almost continually from 1984 through to 1996 (Evans et al., 1996; Silverstone et al., 1996), although they are often characterised as being two waves of reforms, associated with two different governments, The Labour government of 1984–1990 and the National government of 1990–1996. Much has been written about the two sets of reforms (Evans et al., 1996; Silverstone et al., 1996; Singleton, 2006) so we need not discuss these issues in further detail here. It is sufficient simply to say that New Zealand was transformed from a rather highly regulated (Easton, 1994) mixed economy to a highly

deregulated free market economy. Such deregulatory policies were introduced in the belief that these would promote a transformation in New Zealand from an economy with a deteriorating performance to one with an accelerating performance (Shallcrass, 2007). Although similar transformations were taking place in other countries such as Ireland and the UK, in many ways the adjustment process was even more radical in New Zealand (Evans et al., 1996), as it allowed for fundamental reforms to both the central banking system (Singleton, 2006) and also all labour contracts (Harbridge, 1993; Harris & Daldy, 1994). These reforms are generally assumed to have determined the subsequent development of the economy. In particular, in the wake of these reforms, the long-run downward trajectory of New Zealand's labour productivity performance relative to other advanced OECD countries over the previous three decades was arrested.

Since the reform period of the 1980s and early 1990s, in some ways the economy of New Zealand has performed well by OECD standards. Macroeconomic stability has been maintained and, over the last decade, unemployment has remained consistently amongst the lowest in the OECD (OECD, 2008a), and inflation has been kept low (OECD, 2008a; Singleton, 2006) by OECD standards. New Zealand is now a relatively very open economy with a largely transparent institutional system, the third strongest property rights in the world and the world's strongest investor protection environment (Porter et al., 2007), the seventh most flexible labour market in the world and the world's lowest labour firing costs (Porter et al., 2007), the world's lowest number of procedures required to start a business (Porter et al., 2007), high firm competition and turnover (Law & McLellan, 2005), and one of the world's most transparent and least corrupt institutional environments in the world (Porter et al., 2007). As such, New Zealand has relatively light levels of regulation by OECD standards (Gerardin & Kerr, 2003; Hall & Casey, 2006). New Zealand also has a relatively small public sector, and particularly so in comparison to other small advanced countries, which tend to exhibit higher relative government costs due to a lack of economies of scale (Alesina & Spolaore, 2005). New Zealand's total tax revenue as a percentage of GDP ranks 15th out of 30 OECD countries, and is lower than for all but one of the small OECD countries (OECD, 2008a). Meanwhile, New Zealand's relative share of social expenditure ranks only 21st highest in the OECD (OECD, 2008a). It is consistently ranked as one of the world's most liberalised economies,² the second best country in the world for doing business (World Bank, 2009a), the world's lowest levels of corruption,³ the best country in the world for entrepreneurship (Acs & Szerb, 2009), the ninth best for overall institutions (World Economic Forum, 2007), amongst the world's lowest trade barriers (MED, 2005), and one of the best locations for international capital investment.⁴

Yet, in spite of all of the very good macroeconomic and institutional characteristics of the New Zealand economy, there is, however, one apparent anomaly. Although the relative downward slide in the New Zealand's labour productivity performance compared with other countries was arrested in the wake of the economic reforms, there has subsequently been no strong evidence of any real long-run catching-up phenomenon. Structural diversification began in the 1960s (Easton, 1994) and accelerated in response to the reforms of the 1980s and 1990s (Conway & Orr, 2000), as did New Zealand's exports of higher value-added manufactures and services. However, the institutional changes do not subsequently appear to have fostered the levels of diversification into new trade and growth sectors that would be sufficient to generate a catching-up process. During the same

period of time, the industrial structures of many other countries, such as Finland, Ireland, Spain, Korea, Taiwan, the Netherlands, the UK, Denmark and Sweden – all of which to varying degrees are less deregulated than New Zealand and none of which have superior institutions to New Zealand – have managed to adjust in a more successful manner.

Between 1985 and 2003, the 11-year moving average of annual productivity growth in New Zealand was consistently below that of both Australia and also the OECD average, and since 1990 the gap has consistently been between 0.75% and 1.00% (NZIER, 2006a). After 1993, New Zealand's total factor productivity increased from the 1980s levels (Harrington, 2005) and during the period 1992–2002 New Zealand grew faster than most advanced OECD countries (OECD, 2003), with almost half of productivity growth since 1988 being associated with increased labour quality, and half with increased participation (Szeto & McLoughlin, 2008). However, even though growth has improved slightly, none of the New Zealand's major sectors has experienced a substantial increase in total factor productivity since the mid-1990s (Harrington, 2005). Of particular concern is the fact that technical progress has played little role in the growth of NZ industry (Fox, 2005), with productivity largely driven by some of the highest levels of labour utilisation (MED, 2005) and hours worked in the industrialised world (NZIER, 2006a). The result is that today, the GDP per capita productivity levels of New Zealand are still currently only 73% of the Australian level, 82.3% of the OECD average and 82.6% of the average for the 15 western European EU countries (OECD, 2008a). For this reason, although New Zealand always scores highly on its institutions and policies (MED, 2005), its overall competitiveness ranking is still only 24th in the world (Porter et al., 2007), which, as we will see below, is largely consistent with its productivity ranking. As such, instead of observing a classic U-shaped adjustment process in which rapid and fundamental structural adjustments cause enormous disruption, the eventual benefits of which take time to be realised, the New Zealand economy until now appears to have exhibited something of an L-shaped adjustment process, in which there has been no subsequent relative improvement in New Zealand labour productivity performance in comparison to its competitor countries. This is New Zealand's apparent productivity paradox.

To get a sense of this paradox we can consider the relative productivity and income per capita ranking of New Zealand in comparison with the other major advanced OECD economies plus Israel, Taiwan, Singapore and Hong Kong⁵ on the basis of data from the OECD (2008a), Maddison (2006), the Penn World Tables 6.2, and the World Bank (2008a). Comparing New Zealand with all other major advanced economies across a range of different data sources allows us to see the numbers in context, and to avoid being focused on the pros and cons of the construction of a particular index. The relative GDP per capita and income per capita rankings for New Zealand at various points in time and according to the different data sources are given in Table 1 and as we see, they all tell a consistent story.

As we see from Table 1, during the second half of the 1980s, there appears to have been a significant fall in New Zealand's relative productivity position, falling three or four places in the rankings of rich countries during this period. This corresponds broadly to the period of the first wave of economic reforms, a period when the NZ economy experienced little or no economic growth. The five-year moving average annual growth rate for the years 1984–1999 up to and including

Table 1. NZ labour productivity rankings.

NZ rankings	1985	1990	1995	2001	2004	2005	2006	2007
Maddison	19	21	21	22	–			
Penn WT 6.2	17	21	21	23	22			
OECD	–	–	22	22	23	24	24	
World Bank (Income per capita PPP)						25	24	25
World Bank (Atlas Methodology)								24

1988–1993 were all zero or below zero, and only from 1994 onwards was the five-year moving average annual growth rate for New Zealand positive (MED, 2005). Between 1990 and 1995, which is roughly the period of the second wave of reforms, or over the longer period 1990–2001, there was also a very gradual decline in the relative productivity ranking of New Zealand, which fell by a couple of positions during this period, and the process has continued up to today.

According to the Maddison (2006) estimates during the period 1985–1990, which broadly corresponds to the first wave of NZ economic reforms, New Zealand was overtaken by both Hong Kong and Singapore, while the Penn World Tables 6.2 also include UK and Italy as overtaking New Zealand. During the period 1990–1995, which broadly corresponds to the second period of New Zealand's reforms, according to Maddison (2006), New Zealand was not overtaken by any country, while in the Penn World Tables 6.2 New Zealand temporarily overtakes Finland, after the Finnish banking and financial crisis 1990–1993. Subsequently, during the period 1995–2001, which was after the two major waves of New Zealand's economic reforms, New Zealand was overtaken by both Finland (Penn World Tables 6.2) and Ireland (OECD, 2008a; Penn World Tables 6.2). Most recently according to the World Bank (2008a) and OECD (2008a) estimates, during the period 2000–2006, NZ was also overtaken by both Spain and Greece.⁶

It might be argued that it is problematic to use changes in the relative productivity rankings of New Zealand as an indicator of economic performance, because rankings are ordinal measures that do not allow for changes in the distribution of the actual indices. Moreover, this could be particularly problematic here because one notable feature of the productivity statistics of the advanced economies is a tendency towards bunching, such that ranking measures might imply a loss of information. On the other hand, however, this tendency towards productivity bunching itself also highlights an important issue for New Zealand. Most of the debates that take place within New Zealand naturally tend to focus on the economic performance of New Zealand in comparison with its nearest neighbour, Australia. Australia now has much higher productivity levels than New Zealand, and the result of this is that discussions about New Zealand's economic strategy often end up focusing on differences in the minutiae of Australian and New Zealand government policies or tax regimes. Yet, a much wider perspective is required here, in part because the productivity effects associated with the minerals boom is largely specific to Australia (OECD, 2006b), and also because the issues are actually much broader than comparisons between New Zealand and Australia suggest. In 1984, the productivity levels of New Zealand were little different to those of Australia, Canada, and also the productivity levels of almost all western European economies (Maddison, 2006). It is only from 1985 onwards that the productivity levels of New Zealand began to fall behind the productivity levels of other advanced

OECD economies. Between 1993 and 2002, New Zealand's annual growth outperformed the overall OECD growth average (IMF, 2004), and 2001 to 2004 was a period during which New Zealand's five-year moving average growth rate was higher than at any period since the 1960s (MED, 2005). Yet, while the productivity gap in 1993 between New Zealand and the average for the OECD as a whole was 16.2 percentage points, in 2006 it was 17.6 percentage points (OECD, 2008a). What appears to have happened is that during this period, the productivity gap widened very slightly between 1993 and 1997 (OECD, 2008a), closed again slightly until 2002 thanks to a period of strong growth (OECD, 2005), and then since 2003 the productivity gap has slightly widened again as New Zealand's growth has slowed, especially since 2005 (OECD, 2007a). However, even allowing for fluctuations in New Zealand's growth rate, over these post-reform years the variation in the productivity gap has been very minimal. For the last 15 years or so, the productivity performance of New Zealand has varied between 82% and 84% of the OECD average. In terms of income per capita, the gap between New Zealand and its competitors is not only greater than that for GDP per capita, but has also been steadily widening. Since 1993 the income per capita gap with Australia has increased by 7 percentage points, while that with the UK has increased by 11 percentage points. As such, in spite of some recent optimism (OECD, 2005),⁷ whether we employ GDP per capita or income per capita indices (OECD, 2008a; World Bank, 2008a), there is no obvious sign that New Zealand is experiencing any real long-run catching-up process (OECD, 2008a). As such, the productivity gap that first emerged around 1985 and then increased through the reform period of the second half of the 1980s, appears to have remained almost intact for most of the last two decades. Today, the only advanced non-transition countries that are currently still below New Zealand in the productivity rankings are Israel, Taiwan, South Korea and Portugal, and many of these countries are catching up with New Zealand rapidly. On some measures, even the income per capita level of post-transition Slovenia, a very small but not at all geographically isolated country, is now slightly above New Zealand.⁸ In effect, therefore, the productivity gap between Australia and New Zealand is actually indicative of a long-term productivity gap between New Zealand and almost all of the other advanced economies. While the observed high degree of productivity bunching between advanced countries is consistent with the convergence conclusions expected from increasingly liberalised trade and factor markets (Borts & Stein, 1964; Barro & Sala-i-Martin 1992), the performance of New Zealand appears to be an exception.

The commonly-held aspiration for New Zealand to climb into the top half of the OECD productivity rankings,⁹ a position it held approximately in 1977 (Maddison, 2006), implies that New Zealand will need to overtake at least five or six countries and this involves closing a productivity gap of some 21% (OECD, 2008a). However, to get a sense of the scale of the task at hand, at current relative growth rates exhibited by New Zealand and other OECD countries it will take something of the order of two to three decades for New Zealand to achieve its stated target of reaching the top half of the OECD.¹⁰ To catch up more prosperous countries, the productivity gap that needs to be closed is far greater. For example, New Zealand's GDP per capita is currently some 30% below Canada and the Netherlands, 27% below that of Australia, and even some 19% below the two OECD economies which, during the 1990s, experienced the most severe structural difficulties, namely Japan and Germany (OECD, 2008a). At current growth rates, therefore, closing the

productivity gap between New Zealand and countries such as Canada, Australia, the Netherlands and other western European economies, will take three decades or more. In terms of other small advanced economies against which New Zealand is often compared, such as Ireland, Denmark and Finland, the labour productivity gap is 37%, 27% and 21%, respectively. Once again, closing these productivity gaps will take three or four decades at current New Zealand growth rates. If we use income per capita as the index of performance, which allows for inflows and outflows of repatriated profits, then the gap between New Zealand and its major advanced industrial competitors is even greater. This reflects the fact that while New Zealand is very open for inward FDI, its outward levels of FDI are the lowest of any advanced non-transition economy (OECD, 2008a). As such, a higher percentage of GDP is repatriated overseas and a lower percentage of overseas-earned profits is repatriated back to New Zealand, than for almost any other developed economy.

These arguments now allow us to recast the apparent productivity paradox of New Zealand in a slightly different light. If the long run relative productivity performance of New Zealand is L-shaped, then we clearly have a productivity paradox. Why is it not U-shaped? Alternatively, even if we will eventually observe a U-shaped labour productivity adjustment process (and the jury is still out in this), the question still remains: why should the adjustment process have taken something of the order of half a century from the original reforms to achieve this outcome? Half a century is almost certainly a timescale that was never envisaged by the 1980s reformers. New Zealand's productivity paradox may therefore not necessarily be a matter of trajectory, but certainly it is a question of time. In other words, why is it that such radical economic reforms, which ensured, at least in a textbook sense, that all of the requisite institutional aspects were in place for promoting strong economic performance,¹¹ have taken so incredibly long to (hopefully) produce the desired outcomes? This indeed is New Zealand's productivity paradox.

While almost all observers are in agreement that the economic reforms of the mid 1980s to mid 1990s mark a defining moment in the economic history of New Zealand, how these reforms are viewed therefore differs markedly according to how one accounts for the productivity paradox. Much of the current debates between the 'greater-deregulation' camp and the 'greater-intervention' camp therefore largely arise from disagreements as to the exact nature of the long-run impacts of the reforms and the likely L- or U-shape of the adjustment process, with commentators' views ranging from being pessimistic (Dalziel, 2002; Gould, 2008), sanguine (Singleton, 2006), to positive and optimistic (Shallcrass, 2006). The problem with these arguments, as with most attempts at explaining New Zealand's productivity paradox, is that they are all essentially institutional in nature. This is not surprising, given that that the political economy of New Zealand for more than a decade was dominated by fundamental institutional change. Yet, the institutional changes in New Zealand were widely regarded internationally as a textbook set of changes that would foster growth. The fact that this is not what happened means that now even strong advocates of the economic and institutional reforms themselves disagree on the reasons for the paradox, with some suggesting that the reforms were too slow and insufficiently far-reaching, while others argue that they were too fast (Kasper, 2002). In reality, resorting to institutional explanations of New Zealand's productivity paradox gets us nowhere, precisely because there are no institutional models that can account for the apparently paradoxical combination of features of New Zealand's poor economic performance.

This paper, however, takes a very different view of these phenomena. The argument to be developed here is that the currently very slow (if at all) catching-up processes exhibited by New Zealand are nowadays largely unrelated to the arena of New Zealand's past or present domestic policy-making or institutional changes. Neither are they related to the greater-intervention versus the less-intervention debates, or to institutional arguments regarding the past pace of deregulation. Rather, New Zealand's desperately slow productivity adjustment processes are nowadays primarily related to external, enormous, and fundamental changes in the global marketplace, which began to take place between the 1960s and 1980s, but accelerated rapidly since the 1990s. The outcomes of these changes are that, whereas history dealt New Zealand a very good hand for competing internationally in previous eras, history has dealt New Zealand a very poor hand for competing in the current global marketplace. The reasons for this are primarily related to economic geography.

In order to consider why this is so it is much more useful to compare New Zealand's characteristics, behaviour and performance in comparison with that of other advanced economies, rather than with respect to its own previous performance, as is normally done in domestic policy debates. In these terms New Zealand is a rather unusual economy. For an advanced industrialised country it has a very particular mixture of low population scale and low population density (Statistics New Zealand, 2005), high levels of urbanization, high levels of international openness, the most extreme geographical isolation and the lowest market potential of any developed country (OECD, 2008b), a very high reliance on land-based exports, and the lowest level of export diversity for any advanced economy (World Bank, 2008b). Only Iceland and Australia exhibit similar geographical characteristics to New Zealand, although Iceland is a much less geographically isolated economy, and Australia is a very much larger and more diverse economy. Greece also exhibits many similar features to New Zealand in the sense that exports are dominated by agriculture and tourism and the country's economy is increasingly dominated to a very high degree by one urban area. Yet Greece is also very much less geographically isolated than New Zealand (OECD, 2008b). Evidence for the role that geography plays in economic performance comes from the World Bank (2009b) and the OECD (2008b,c), both of which argued that the adverse effects of geographical isolation on labour productivity are greater in the case of New Zealand and Australia than for any other advanced economy. OECD (2008b,c) estimates for both countries suggest that distance reduces labour productivity by more than 10%. IMF (2004) estimations also imply that some 50% of the labour productivity gap between New Zealand and the OECD average can be attributed to geography alone. These IMF (2004) results are almost exactly consistent with OECD estimates, which suggest that just over 10 percentage points (OECD, 2008b,c) of the 21 percentage point gap (OECD, 2008a) between New Zealand and the OECD average can be attributed to geography alone. However, geographical isolation may not be the whole story. One major advantage that much larger countries such as Australia have over New Zealand is the widespread existence of economies of scale (NZIER, 2006a) associated with a larger domestic market and larger diverse cities. Indeed further estimations (OECD, 2008c) suggest that Australia's scale more than compensates for the adverse effects of distance, whereas in the case of New Zealand, the small country effect means that the productivity gap increases to more than 14%. While Ireland, Sweden, Denmark, Norway, Finland,

Austria, Singapore and Hong Kong are also all small economies with population levels of a similar scale to New Zealand, none of these countries is in any way geographically isolated from large markets in the way that New Zealand is (OECD, 2008b), and all of these countries have much more diverse export profiles than New Zealand (World Bank, 2008b). As such, there is much evidence to suggest that the problems of achieving economies of scale and scope also appear to be related to achieving accessibility to major markets, and being a small country that is immediately adjacent to large markets allows for the realisation of such economies of scale. In contrast, being small and isolated limits the ability of a country to achieve economies of scale. These complex interrelationships between economies of scale, production diversity, location and trade are the domain of economic geography.

Following on from The Treasury's Economic Transformation Project a small number of commentators within the New Zealand government ministries (Crawford, 2004; Lewis, 2002; Procter, 2008; Simmons, 2004; Skilling, 2001a,b) and various other public policy think-tanks such as the New Zealand Institute (Skilling & Boven, 2005a,b, 2006a,b) and the Lowy Institute (Gyngell et al., 2007) have for a long time been arguing that economic geography explanations for New Zealand's poor productivity performance should be considered seriously, as have commentators within Australian government ministries (Davis & Ewing, 2005). It will be argued in this paper that the basic insights of this small number of commentators are correct, because economic geography points to particular types of interrelationships between location, scale, productivity, and diversity, all of which succinctly characterise New Zealand's current predicament. These are therefore the issues to which we now turn.

3. The contemporary characteristics of globalisation

New Zealand has always been where it is, and for many years, as we have seen, New Zealand was one of the world's richest countries. Therefore, some commentators such as Kasper (2002) regard the idea that economic geography can explain New Zealand's current productivity paradox as a 'view [that] can be quickly dismissed', and that the 'excuse of isolation and small size is not credible; the record suggests that it is the quality of institutions which societies adopt that matters for long-term growth.' In reality, commentators such as Kasper have completely missed the point. Kasper's views come from 1980s' and 1990s' thinking in development economics, but these arguments always faced problems when applied to many of the advanced economies, whose productivity performance varies little yet whose institutional structures vary enormously. Moreover, as we have already seen, the quality of New Zealand's institutions is second to none, so while these arguments may have had credibility in the 1980s, their analytical power in accounting for New Zealand's performance today is extremely limited. Moreover, many of the institutional arguments have been shown recently to have only limited applicability even when applied to developing countries (Rodrik, 2007). Indeed, during the last decade, the international consensus in development economics has changed dramatically, and nowadays points to the critical role played by economic geography and agglomeration in explaining differences in the contemporary economic performance of countries (Collier, 2006; World Bank, 2009b). This does not serve to rule out the importance of institutions, but to add a much broader perspective to the argument.

The economic geography of the world economy has changed fundamentally over the last two decades, and the reasons for this changes are related to the

characteristics of the most recent phase of the long-run process of globalisation (Easton, 2007). As we will see below, the nature and scale of spatial transactions costs have changed radically, and the evidence regarding the contemporary role of agglomeration economies in driving productivity growth is now so overwhelming that it is more or less beyond question (Venables, 2006). However, both the theoretical and empirical evidence in support of the following arguments is so extensive that it is simply not possible to discuss it in detail here. Therefore, for reasons of brevity, the reader is referred to various publications (Fujita et al., 1999; Fujita, 2007a; Henderson & Thisse, 2004; Krugman, 2007; McCann, 2008, 2009; OECD, 2006a, 2007d; World Bank, 2008b). In terms of the issue at hand, as we will also see shortly, the result of the reshaping of global economic geography is that while New Zealand's cartographical location has not changed, in terms of economic geography New Zealand's position in the global economy has indeed changed. It is this change that largely accounts for the productivity paradox of New Zealand.

In order to begin to understand this, we need first to consider the major characteristics of the current era of globalisation and then consider how global economic geography is being shaped by these characteristics. Then we can apply these arguments to the case of New Zealand.

Globalisation processes have been continuing for approximately four centuries, driven at each stage by different technological and political developments. Yet, much of the 20th century can be seen as an exception to these long run processes. The catastrophes of the two world wars and the inter-war Depression lead to massive relative contractions of global trade and international investment (Findlay & O'Rourke 2007; McCann, 2008). The ensuing Bretton-Woods system insulated individual national economies from many aspects of globalisation processes, the result of which is that the global trade-to-GDP ratios of 1929 were not again realised until 1950, and the global assets to GDP ratios of 1914 were not achieved again until 1980 (Crafts, 2004). In Europe, the result of the founding of the EEC in 1958 and EFTA in 1960 was that the European nations experienced a rapid growth in cross-border trade, investment and competition during the 1960s. The impact of these first steps of regional economic integration was that European per capita incomes in OECD countries were boosted by between 4 and 8% between 1959 and 1969 (Eichengreen, 2007). US foreign direct investment (FDI) in Europe also accelerated dramatically during the 1960s, and the resulting technology transfer arising out of the trade and investment complementarities acted to spur European technological growth (Eichengreen, 2007). The emergence of the Eurodollar markets also facilitated similar technology transfer mechanisms from US to European service industries during the 1960s, and the deregulation in the global capital markets in the 1970s and 1980s, along with the rise of new Japanese and Korean manufacturing giants, set the stage for entirely new interrelationships between countries by the late 1980s. Yet, in many ways, the most fundamental changes in the global economy all happened within a period of less than five years, between the end of the 1980s and the early 1990s, and these changes were both technological and institutional in nature (McCann, 2008, 2009).

In terms of technological changes, the 1960s to the 1980s saw the rapid rise in modern transportation and communications technologies. New forms of freight haulage developed around the principles of containerisation, and roll-on roll-off technologies allied with GPS satellite systems burgeoned for all forms land and sea transportation. At the same time, air transportation became ever more affordable as

technological improvements were associated with massive deregulation in global transportation systems and networks (Combes & Lafourcade, 2005). Meanwhile, the 1980s saw the emergence of widespread use of computing in business, and the emergence of mobile telephone technologies. However, the advent of the internet in 1991 allowed these diverse technologies to become increasingly integrated for the first time. Global information and communications technologies (ICTs), as we nowadays understand them, are only a very recent phenomenon, and are basically a product of the 1990s merging of computer and telecommunications technologies (McCann, 2008).

At the same time as these fundamental global technological changes were taking place, equally fundamental global institutional changes were also taking place. International integration and trade localization processes had been progressing steadily in Europe between the 1960s and the 1980s. However, the late 1980s onwards saw the start of fundamental global institutional changes. The fall of the Berlin Wall in 1989, the start of the market reforms in China after 1989, the 1989 industrial reforms in Indonesia, the re-admittance of South Africa to the global community after the release of Mandela in 1990, the 1991 deregulatory reforms in India (Kambhampati & McCann, 2007), and sweeping market reforms and the new currency in Brazil in 1993, together brought some 1.5 billion new workers into the global economy between 1989 and 1993 (Venables, 2006). At the same time, the creation of the EU single market in 1992 and the formation of NAFTA in 1994, and further integrating developments in the South and East Asia region led to the development of three super-regional areas of economic integration built around enormous internal markets and free trade areas. The development of these internal markets and free trade areas have also been associated with rapid rises in the number of bilateral investment treaties and double taxation treaties, the number of which is very closely associated with geographical proximity between the partner countries (McCann, 2009).

Finally, in response to the commercial opportunities afforded by these technological and institutional changes, over the last two decades we have recently witnessed an enormous increase in organizational changes, whereby firms out-source and 'off-shore' increasing numbers and types of economic activities to the newly-emerging economies.

Much of the popular public perception of globalisation is increasingly that the 'world is flat' (Friedman, 2007), which implies that globalisation is primarily about firms from advanced countries taking advantage of the new technological and institutional opportunities in order to shift increasing numbers of activities to developing countries. Yet, the world is far more complex than this. Relative to global GDP, the share of global inward and outward FDI, global trade and global R&D is increasing. However, over the last two decades, the share of global output, global trade, global foreign direct investment and global R&D, which is accounted for by all three of the super-regions (EU, NAFTA and South and East Asia) has increased (Fujita, 2007b,c; McCann, 2008, 2009). The reason why increasingly localised economic integration is associated with increasing output, is primarily because of the increasingly important role played by cities in generating agglomeration economies. The importance of cities in shaping not only the spatial distribution of activity but also the spatial distribution of productivity is now regarded as basically being beyond question (OECD, 2006a; Venables, 2006; World Bank, 2008b). The result of these agglomeration economies is that many aspects of the current phase of

globalisation can more accurately be described as ‘global regionalism’ (Rugman, 2005), whereby countries within the same individual super-region are becoming more closely integrated with respect to one another, relative to other countries. As such, in terms of economic geography, almost all of the empirical evidence suggests that the world is not at all becoming more flat (Friedman, 2007), but rather is becoming far more uneven (McCann, 2008). The reasons for the increasing unevenness in global economic geography can be identified by considering recent changes in the nature of spatial transactions costs and relating these to the changing role of agglomeration economies.

4. Some economic geography insights

There is a vast literature on economic geography and urban economics (Clark et al., 2000; Fujita et al., 1999; Fujita & Thisse, 2002; Henderson & Thisse, 2004; Krugman, 1991), but it is important for our analysis to highlight some of the fundamental theoretical insights of these fields, and then to relate them to empirical observations. In particular, it is necessary to take on board the major insights regarding the relationship between agglomeration and spatial transactions costs, and secondly the insights regarding the role of agglomeration itself.

In terms of the relationship between agglomeration and spatial transactions costs, we can consider a situation where two economies produce two sets of outputs, one under conditions of constant returns to scale, and the other under increasing returns to scale. In terms of our discussion here, we can regard low knowledge-intensive and low value-added goods as those produced under broadly constant returns to scale, while high knowledge-intensive and high value-added goods are those produced broadly under (agglomeration) economies of scale. Under these assumptions it can be demonstrated (Krugman & Venables, 1995; Fujita et al., 1999) that with extremely high spatial transactions costs all countries and regions have similar production patterns, similar levels of productivity and similar levels of wages. The reason is that the high spatial transactions costs act as trade barriers, so local production provides for local markets. However, as spatial transactions costs fall in an environment of increasing returns to scale, if labour is mobile, a centre–periphery divergence forms in which the central regions exhibit agglomeration effects. Only at extremely low levels of spatial transactions costs, which asymptotically approach zero, does the spatial economy revert to an even distribution of activity. As such, the relationship between agglomeration and spatial transactions costs therefore exhibits an inverse U-shape.

As spatial transactions costs fall, the regions experiencing the agglomeration effects exhibit higher output, higher productivity, higher wages and higher land prices, as well as a greater diversity of production, labour skills and inflows of labour. Peripheral regions, in contrast, exhibit lower output, lower productivity, lower wages, lower land prices, a lower diversity of production and skills, and outflows of labour. Moreover, the divergence between the core and periphery regions is magnified if the peripheral regions are also small with only small urban centres at their core. This argument can be applied more or less equivalently to the case of regions within the same country, or to the case of geographically neighbouring countries where there is a high degree of mutual openness and international labour mobility. Only when spatial transactions costs approach zero does an even distribution of economic activity once again appear.

In terms of the role of agglomeration itself, there are many arguments as to why agglomeration may be advantageous. Agglomeration increases competition via the increased mutual transparency and visibility associated with geographical proximity (Porter, 1990). In some cases it also facilitates the networking and coordination between firms (McCann, 2003). Agglomeration provides a nursery role for new enterprises (Duranton & Puga, 2001) and agglomeration also reduces risks by improving labour market matching processes. However, the major advantage of agglomeration in the modern world is widely regarded as being about maximising the efficiency and effectiveness of knowledge exchanges (Glaeser, 2008) in the production processes of high value-added goods and services in a diverse economic environment (OECD, 2006a). This is quite different in nature to previous eras when industrial agglomeration in cities was primarily associated with the minimisation of transport costs within highly specialised centres (Glaeser & Kohlhase, 2004).

The arguments outlined here imply that if globalisation is associated with falling spatial transactions costs and agglomeration economies, then globalisation will lead to increasing differences between the fortunes of regions and cities within the same parts of the global economy and even within the same country. In terms of empirical observations there is a great deal of evidence which suggests that the current era of globalisation has indeed been characterised by changes in spatial transactions costs and agglomeration economies, which are largely consistent with the theoretical arguments described above.

In terms of spatial transactions costs, the spatial transactions costs associated with low knowledge, low value-added, standardised and relatively routinised activities, have unquestionably fallen dramatically over recent decades. The transport-movement costs have fallen by some 95% over the last century (Glaeser & Kohlhase, 2004), and telecommunications costs have also fallen by a similar order of magnitude over just the last 35 years (OECD, 2008b). As such, the geographical dispersion of low knowledge, low value-added, standardised and relatively routinised activities becomes ever easier, and this accounts for the 'world is flat' arguments of Friedman (2007). In contrast, the spatial transactions costs associated with high knowledge, high value-added, non-routine and non-standardised activities have actually increased over recent years. This is because of the increasing importance of timeliness, speed, variety, customisation, and service-quality, in all high knowledge-intensive forms of production¹² and service delivery (Disdier & Head, 2008; Duranton & Storper, 2007; McCann, 2007). The principal reason for this is that the premium associated with face-to-face contact in high knowledge intensive activities appears to have increased (Gaspar & Glaeser, 1998; McCann, 2008; Storper & Venables, 2004), because the spatial transactions costs on the inputs side of the production process, rather than the outputs side, have increased for high value goods (McCann, 2008). Moreover, the increased importance of face-to-face contact has taken place precisely at the same time that information and communications technologies (ICTs) have been improving dramatically. The reason appears to be that increased ICT usage implies that more complex and sophisticated levels of knowledge are being generated, analysed and manipulated, at an ever-increasing pace, the result of which is that product life-cycles are continuously shortening and customisation increasing. The trust and confidence required for the successful management and transmission of this knowledge in this environment therefore requires even more intense and frequent face-to-face contact (Gaspar & Glaeser, 1998; McCann, 2008; Storper & Venables, 2004). The increased frequency of

face-to-face contact increases input spatial transactions costs (McCann, 2007). It is no accident that the sectors that use ICTs the most intensively, such as the electronics and semiconductor industry and the international financial services sector, are amongst the most spatially concentrated industries in the world, clustering in places such as Silicon Valley, Wall Street and the City of London's Square Mile.

The increased spatial transactions costs for knowledge activities and the resulting premium for face-to-face contact for high knowledge inputs also accounts for the increasing importance of cities and agglomeration in the current era of globalisation. Prior to the 1980s, major cities were actually in decline in many parts of the world (Glaeser, 1997) as firms and people increasingly relocated to smaller centres. Yet, the advantages of scale and diversity have re-emerged with a vengeance over the last two decades in almost every part of the world. Indeed, there is overwhelming evidence that economic growth and globalisation over the last two decades has favoured larger urban centres in almost every country (Brakman & van Marrewijk, 2008; World Bank, 2009b). Over recent decades, cities dominated by a diverse range of knowledge-sectors have tended to perform better than cities that are highly specialized (Glaeser et al., 1992; Combes, 2000), and cities with a highly diversified range of knowledge-sectors also tend to be bigger than the highly specialised cities (Henderson & Thisse, 2004). Over recent decades, major cities have tended to be the cities that most benefit from the increasing scale advantages associated with being nodes in global airline and trade networks (McCann, 2008). Over recent decades major cities are also increasingly the locations towards which high quality human capital migrates (Berry & Glaeser, 2005; Faggian & McCann, 2009). The results of all of these changes are that cities nowadays exhibit the highest levels of productivity (Ciccone & Hall, 1996; OECD, 2006a), the highest levels of innovation (Carlino et al., 2007), the highest levels of entrepreneurship (Acs, 2002; Acs et al., 2008), and the relative importance of cities to national competitiveness and global output is increasing all over the world (OECD, 2006a; World Bank, 2008a). The evidence therefore suggests that there are major advantages associated with industrial clustering and agglomeration for high knowledge-intensive and high value-added activities, and that the geographical concentration of these types of activities is becoming ever more important.

A related issue here is the relationship between cities, areas of economic integration, and role of multinational firms. The forces of agglomeration described above are increasing most noticeably in areas of economic integration. The reason is that these areas allow for the increased mobility of factors, and the more that factors are mobile the faster are the agglomeration processes. Yet, not all cities are benefiting from these processes, because in the current era of globalisation, it also depends on the role played by multinational firms (McCann & Acs, 2009). The world's most productive cities (OECD, 2006a) are also the world's most globally connected cities in terms of the scale and variety of their corporate and transport networks (McCann, 2008, 2009; McCann & Acs, 2009). As such, the cities that perform the best in the current era of globalization are those cities that are located in an area of increasing economic integration and that are also highly globally connected.

5. Cities, globalisation and New Zealand's productivity paradox

We can relate the above economic geography arguments to the case of New Zealand at two rather different levels, both of which take place simultaneously. First, we can

consider these arguments at a super-regional international scale, and second we can consider these arguments at a sub-national inter-regional level.

In terms of a super-regional international scale, the relationship between New Zealand and Australia since the foundation of CER in 1983 is in many ways evolving from an international trading system to an inter-regional system, whereby countries that were previously characterised primarily by fixed factor endowments and significant mutual trade barriers are increasingly characterised by highly mobile factors and minimal mutual trade barriers. Evidence for this comes from the fact that Australia now accounts for one fifth of New Zealand's exports, over half of New Zealand's outward FDI stocks, over one half of New Zealand's inward FDI stocks, and New Zealand is Australia's fifth largest export market (NZIER, 2006a). Moreover, Australian FDI stocks in New Zealand doubled between 2001 and 2005 while New Zealand's stock of FDI in Australia tripled between 2001 and 2005 (NZIER, 2006a). In terms of labour, Australia is also by far the largest overseas residence destination for New Zealand citizens. The number of New Zealanders living in Australia is between six and eight times the number of New Zealanders living in the second largest destination of expatriate New Zealanders, which is the UK (Dumont & Lemaitre, 2005).¹³ We can contrast these figures with the fact that, in 1965, just 4.7% of New Zealand's exports went to Australia (Gould, 1982).

In theoretical terms, the emerging relationship between New Zealand and Australia is akin to two previously largely-closed economies in a new economic geography framework (Fujita et al., 1999; Krugman & Venables, 1995), which then undergo rapid integration, and this has very specific economic geography effects. In particular, one economy, namely Australia, exhibits both relatively larger agglomeration economies and also a larger home-market effect than the relatively smaller economy, New Zealand. As the economies become increasingly open to each other with both trade barriers and transport costs falling, the greater industrial diversity of the larger agglomerations in the larger country provides for a greater variety of employment opportunities for mobile workers and a greater variety of input and output linkages for mobile firms. The result is that Australia becomes relatively more attractive for both capital and labour than New Zealand. The increased attractiveness of Australia for capital therefore implies that Australia will increasingly become a more important focus for global and multinational mobile investment than New Zealand. An obvious manifestation of this would be that Australia would be expected to exhibit relatively higher levels of capital stocks per worker than New Zealand, and also an increasing number of higher-order top-level decision-making and corporate headquarter functions than New Zealand. Meanwhile, the increased attractiveness of Australia for labour implies higher wages and increased labour migration flows from New Zealand to Australia. In particular, highly-skilled workers will be particularly increasingly mobile as they seek to take advantage of the wage premia they can command in Australia due to higher capital labour ratios.

In terms of a sub-national interregional scale, largely the same phenomena will be expected to operate within the country as between countries. As such, we would expect that New Zealand's urban population will increase relative to its rural population as the role of cities becomes increasingly important. The most obvious issue here is that in the absence of severe congestion constraints, in the long run Auckland will be expected to continue to grow relative to most other New Zealand regions.

Applying the economic geography arguments to the case of New Zealand provides fairly clear predictions:

- (i) A lack of scale and extreme geographical isolation means that New Zealand's labour productivity falls behind that of its advanced country competitors, and in particular Australia.
- (ii) The labour productivity gap will be associated with increasing capital and labour flows from New Zealand to Australia.
- (iii) Existing differences in urban scale across New Zealand's regions will be accentuated as internal migration flows continue to be orientated towards the largest cities.

There is already much empirical evidence that points to exactly these types of predicted outcomes. We can first consider the relationship between New Zealand and Australia described by predictions (i) and (ii) above, and then we can consider prediction (iii).

Regarding predictions (i) and (ii), in terms of capital stocks, Hall and Scobie (2005) estimated that between 1995 and 2002 some 30% of the difference in the rate of growth of labour productivity between Australia and NZ was accounted for by differences in total factor productivity, with some 70% being due to differences in capital intensity between the countries. Obviously, Australia's booming mining sector will account for some of these capital stock differences, but this argument is not sufficient to explain why there is such a divergence between the two countries. Economic geography, however, does provide an explanation, and in support of the economic geography arguments, the available anecdotal evidence suggests that the last decade has also been a time during which increasing numbers of corporate headquarters and decision-making functions are being moved out of Auckland and into Australia (Gravitas, 2003).

In terms of labour mobility, as of 2001, over 460,000 New Zealanders lived permanently overseas (Bryant & Law, 2004), with some 355,000 estimated to be living permanently in Australia in 2005 (Dumont & Lemaitre, 2005). By the end of 2008 the estimated number of New Zealanders living in Australia was estimated to have risen to 477,000.¹⁴ Meanwhile, just under 60,000 New Zealanders live and work permanently in London (Dorling & Thomas, 2004; Dumont & Lemaitre, 2005). Moreover, as we know from migration models, those who are the most likely to move to high wage locations are highly educated young people (Faggian et al., 2007) and the evidence from New Zealand points exactly to this phenomenon (Smart, 2006). Indeed Australian estimates suggest that up to 80% of 'complex problem solvers' working in highly-skilled and high knowledge-intensive occupations are in the age group 25–40 (McDonald & Temple, 2006), and those migrants with these types of skills are by far the most likely skill group to remain in Australia after they enter into employment than those with low skills (McDonald & Temple, 2006). In contrast, New Zealand, relative to its size, exhibits the highest probability of its talented people moving to other countries for work of any advanced country (Porter et al., 2007) and also the highest relative outflows of university graduates of any country in the world (Birrel et al., 2006).

As long as there are greater agglomeration effects in Australia than in New Zealand, as the costs of travel fall (and here we define travel costs in terms of all the transport, cultural and institutional adjustment costs associated with migration),

increasing numbers of mobile highly educated young New Zealanders are likely continue to migrate to Australia to seek higher wage employment. Indeed, recent empirical evidence suggests that such outflows are not only increasing (Smart, 2006) but are also highly correlated with the productivity gap between the two countries (Poot, 2008, 2009). Moreover, the greater are the private costs of New Zealand's university education relative to New Zealand's wages, the greater will be likely outflows as graduates in particular seek to maximise the returns to their human capital. Preliminary evidence already also points to this conclusion (Smart, 2006). These human capital migration flows can become cumulative in the sense that the greater are the out-migration flows from New Zealand to Australia's cities, the greater are human capital inflows to Australia's cities, and the greater will be their productivity and wages, thereby contributing to greater inflows from New Zealand. Such cumulative patterns of human capital mobility have already been observed within the UK (Faggian & McCann, 2009). Taking account of all of the evidence available, Davis & Ewing (2005) therefore conclude that not only do economic geography arguments provide the most promising explanations of the productivity gap between NZ and Australia, but given all of these issues discussed here, the gap is actually not outside reasonable expectations. Meanwhile, at a higher level and at a greater spatial scale, analyses by the Australian Treasury suggest that similar economic geography arguments also explain much of the productivity gap between Australia and USA (Battersby, 2006; Davis & Rahman, 2006).

Thinking of the economy of New Zealand in terms of modern economic geography theories, we are also led to the conclusion that New Zealand can increasingly be considered to exhibit characteristics that are typical of a region of Australasia whose dominant urban centre is Auckland (Grimes, 2005a; MED, 2007), in much the same way as Australia's states each have a dominant urban centre at their heart of their economy (BITRE, 2009). Indeed, by treating New Zealand as a region of Australia and estimating a rank-size rule model for the whole of Australasia, the results conform to what we would expect from Zipf's Law (Maré, 2008). As such, the size distribution of New Zealand's cities closely corresponds to what would be expected by treating Australia–New Zealand as single interregional system. Even on these criteria, however, Auckland still underperforms in terms of productivity relative to the Australian cities. Adelaide is approximately the same size as Auckland, and similar to Auckland's role in the New Zealand economy in that Adelaide is the biggest economy and also has the highest productivity of any part of South Australia (DTRS, 2004). Yet both Adelaide and South Australia as a whole have higher productivity than New Zealand (BTRE, 2004; MED, 2007; NZIER, 2006a), with Tasmania being the only state of Australia that exhibits lower productivity than New Zealand (NZIER, 2006a). Not surprisingly, the age distribution patterns of labour migration from New Zealand to Australia (McCann et al., 2009) are very similar to the labour migration patterns from Tasmania to the rest of Australia (BITRE, 2008).

Following the economic geography arguments, the cumulative processes in favour of Australia can only be reversed when capital investment begins to flow to New Zealand and away from Australia. Yet, this will only happen when either Australia's urban areas face biting capacity-congestion constraints (Nijkamp & Poot, 1998) or the spatial transactions costs involved in managing fixed capital investments in New Zealand from Australian headquarters become so extremely low that Australian firms will be willing to offset less diversity and scale effects in New

Zealand in return for lower wages in New Zealand. However, it would appear that we are as yet currently nowhere near this scenario.

With regard to prediction (iii) above, as well as operating internationally, the economic geography forces are also operating internally within New Zealand. If we examine the internal spatial structure of New Zealand, the highest per capita productivity region of New Zealand is Auckland (Maré, 2008), which is also the largest agglomeration of New Zealand. The region accounts for approximately just over 33% of the population and just over 40% New Zealand's output (Maré, 2008; NZIER, 2006b). During the decade spanning 1991–2001, Auckland was New Zealand's second fastest growing major labour market area, after Tauranga (Baxendine et al., 2006) and provisional 2006 Census estimates indicated that the regional area with greatest growth rate during the more recent period of 2001–2006 was also Auckland.¹⁵ Auckland's relative per capita productivity, calculated with respect to the national per capita productivity level, is ranked 24th in the OECD (McCann & Acs, 2009), and its levels of entrepreneurial dynamism relative to the national level is ranked the highest in the OECD on some criteria (Acs et al., 2008). Not surprisingly, wages are also the highest in New Zealand's two largest cities of Auckland and Wellington (Lewis & Stillman, 2005, 2007) and the wage gap between Auckland and Wellington and the rest of New Zealand's regions grew sharply during the economic reform period (Karagedikli et al., 2000). Auckland has New Zealand's highest number of headquarter functions associated with the world's major companies (ARC, 2006: Auckland City Council, 2006), and evidence also suggests that increasing numbers of New Zealand corporate headquarter functions have moved out of Wellington and into Auckland (Gravitas, 2003). The result is that the ratio of Auckland's Central Business District (CBD) land prices to those in the outer suburbs increased almost monotonically between 1991 and 2004 (Grimes & Liang, 2008).

These observations all point to the contemporary importance of agglomeration within New Zealand itself. Similar evidence can also be found regarding the increasingly important role played by Australia's major cities in Australian national income (BITRE, 2009; BTRE, 2004) and as an employment location for those with high levels of human capital (Bradley & Gans, 1998). Yet, in spite of New Zealand being a deregulated and institutionally transparent economy, Auckland's productivity is some 15% below that of all the major Australian cities (AREDS, 2008), and only ranks as the 46th highest per capita productivity agglomeration in the OECD (OECD, 2006a). The reason for the lower productivity appears to be related to both scale and connectivity. All the available economic geography evidence suggests that scale matters. Even though there is no simple direct relationship between urban scale and productivity, all of the world's most productive cities are at least twice the size of Auckland, and most are between three and five times the size of Auckland (OECD, 2006a). Moreover, economic geography also tells us that location also matters. Apart from Perth in Australia, New Zealand's cities are the world's most isolated advanced urban economies, the result of which is that no urban agglomeration in New Zealand ranks within the world's top 50 most internationally connected cities (Mastercard, 2008), or within the world's top 50 financial centres (CoL, 2008). Auckland is still some distance from being regarded as a global city (Rowe, 2006), and given its isolation and small national market, it appears to be that Auckland is still not big enough to compete with many overseas cities. With a population of just over 1.2 million it is not even amongst the largest 70 urban agglomerations within

the OECD. Moreover, the fact that a small city is also embedded within a small isolated economy of just over a million people means that Auckland is still not large enough to generate the globally competitive agglomeration advantages of the sort discussed by Porter (1990).

Further evidence for is comes from the fact that New Zealand's innovation performance is even less impressive than its productivity performance, and this poor performance can also be accounted for by economic geography arguments. As we have already seen, New Zealand is a highly entrepreneurial society (Acs & Szerb, 2009), and Auckland is a highly entrepreneurial city by international standards (Acs et al., 2008; Rowe, 2006). Yet, when it comes to translating an entrepreneurial culture into innovation, New Zealand ranks very poorly by international standards. Although New Zealand's innovation probability scores are good (MED, 2005), the World Economic Forum (Porter et al., 2007) ranks New Zealand 27th in the world in terms of its overall capacity for innovation, which is the fourth lowest ranking for any advanced economy. As such, New Zealand's innovation ranking is actually slightly worse than its productivity ranking, and not surprisingly this is also consistent with its multi-factor growth ranking of 27th in the OECD (OECD, 2008a). Following economic geography arguments, the most likely reason for this is that the translation of entrepreneurship into innovation is also very much related to issues of scale and accessibility, because knowledge-generation processes, knowledge flows and knowledge spillovers are all found to be related to scale and accessibility. Economies that are large enough to support very large R&D-intensive networks or organisations are able to generate large numbers of innovations. In some cases these innovation-generation processes will take place primarily within individual large firms whose expenditure on R&D will be very large. In other cases, the scale effect is provided for not by individual firms but by high urban populations (Acs, 2002) and high urban population densities (Carlino et al., 2007). Large and dense cities generally maximise knowledge spillovers between firms of all sizes, including many small firm entrepreneurial start-up firms (Acs, 2002). As such, individual large firms or agglomeration clusters of small firms represent alternative regimes by which knowledge-related economies of scale can be achieved (Iammarino & McCann, 2006; McCann & Arita, 2006).

In the case of New Zealand, just under 50% of the total business R&D in New Zealand is accounted for by firms of less than 50 employees and almost 80% is accounted for by firms of less than 250 employees (OECD, 2007c). In both cases these are the highest relative R&D contributions by small firms of any OECD country. The equivalent numbers for the USA are less than 10% and less than 20%, respectively. Moreover, these figures are not necessarily typical for small countries. The total share of business R&D accounted for by firms with less than 50 employees in New Zealand is twice that of the next highest country, Australia, and five times that of Sweden, Finland and The Netherlands. Similarly, the total share of business R&D accounted for by firms with less than 250 employees is only 20% in countries such as Finland and Sweden, and less than 30% in Austria and Denmark (OECD, 2007c). Yet, the highly skewed contribution of small firms is not compensated for by the presence of a large local market or a large and dense local urban environment. While per capita innovation scores in NZ are much lower than for many other countries, patent data evidence suggests that once city sizes are controlled for, per capita innovation rates in New Zealand are similar to other countries (Hendy & Sissons, 2009). As such, the New Zealand economy appears to have neither the

market scale, the urban scale nor the R&D levels to transform world-leading levels of entrepreneurship into world-leading levels of innovation. While New Zealand's innovation performance is good if we control for market and firm size, distance and sectoral composition (Crawford et al., 2007), the point remains that the New Zealand's geography is a disadvantage, and this is a serious problem for productivity. While 'kiwi entrepreneurship' (Acs & Szerb, 2009), and even 'Auckland entrepreneurship' (Acs et al., 2008) are indeed realities, unfortunately 'kiwi innovation' is a myth. Innovation is a function of complex structural systems including both institutions and economic geography (Acs, 2002; Breschi & Malerba, 2005; Feldman & Massard, 2001; Polenske, 2007) rather than simply a result of personal or cultural attributes and aspirations (Acs & Szerb, 2009). New Zealand's media and politicians would do well to drop the misleading mantra of 'kiwi innovation', so as to allow for serious debates as to the possible way forward in this arena.

The wide-ranging arguments and evidence here suggest therefore that economic geography forces are operating on the New Zealand economy both internationally and also internally. These same economic geography arguments also imply that that in the current era of globalisation, the small scale of New Zealand, its extreme isolation, and its low levels of output diversity (World Bank, 2008a) all serve to greatly limit its national productivity potential. New Zealand flourished during the 19th century globalisation processes, when spatial transactions costs were more than 20 times their current levels (Glaeser & Kohlhase, 2004) and when trade was a carefully managed bilateral process within a colonial system. During much of the 20th century, globalisation processes were also retarded, so once again New Zealand experienced no adverse effects from them. However, from the late 1980s onwards, the return of globalisation processes with a vengeance was characterised by rapidly falling international and domestic spatial transactions costs (OECD, 2008b), increasing mutual openness between countries, increased international labour mobility, and a much greater importance for agglomeration economies. In contrast, the inherited spatial and industrial structure of New Zealand as a small, geographically isolated economy with a relatively low level of industrial diversity, exports which are dominated by agricultural goods and tourism, and low levels of outward FDI (OECD, 2008a), is a combination of attributes that in the present era of globalisation is simply not advantageous. Unfortunately, there is nothing about this particular combination of structural features that would suggest that New Zealand ought to be a relatively high productivity economy. Moreover, this is true irrespective of the degree of flexibility in the domestic labour market, the degree of transparency in the local institutional environment, or the levels of cultural aspirations for success (Callaghan, 2009).

The problem for New Zealand is that the return of globalisation processes took place at exactly the same time that New Zealand was undergoing its own economic reforms, and the structural features that are advantageous in the current era of globalisation are very different from the structural features exhibited by New Zealand. Even after undergoing radical economic reforms, New Zealand's productivity is still relatively low by international standards because, as a whole, it is the wrong type of economy for the current era of globalisation (Mason & Osborne, 2007). Globalisation currently favours other types of places in other locations.

Looking at New Zealand's characteristics in the current phase of globalisation through the lens of economic geography, therefore suggests that the apparent

productivity paradox of New Zealand is really a conundrum, a riddle, with a very simple solution. The major characteristics of the New Zealand economy, i.e. a small and extremely isolated economy, with small urban centres, and a low degree of export diversity, is a combination of structural characteristics that is not productivity-enhancing in the modern phase of globalisation, relative to other countries in other places. As such, the observed productivity performance of New Zealand is more or less exactly as expected according to economic geography theory. Moreover, many other characteristics of the relationship between New Zealand and Australia and between New Zealand's own regions internally are also largely as predicted by economic geography. On the principle of Occam's Razor, economic geography therefore provides a much more powerful diagnosis of New Zealand's current productivity performance than many other institutional, structural or macroeconomic arguments. The reason is that although New Zealand underwent fundamental institutional reforms in the 1980s and 1990s, at exactly the same time as this was taking place the landscape of global economic geography was shifting in favour of other places. It may well be that the deregulatory reforms limited some of the most adverse aspects of these shifts, thereby minimising the productivity gap. Yet the point still remains that the world changed, and the world of the late 20th and early 21st centuries is very different from the world that provided New Zealand with almost a century and a half of productivity advantages (Crocombe et al., 1991).

6. Policy issues

One of the implications of these economic geography arguments is that, to a large extent, the ability of individual national governments to effect major changes in their domestic economies is very limited. This is particularly so in the case of a small country such as New Zealand, which seeks to raise its relative productivity levels. The reason is that when we are considering relative productivity between countries, New Zealand's productivity drivers are primarily global external drivers, and as such they respond very little to specific regulatory changes, changes in various tax treatments, or changes in labour regulations, in any one particular country. However, acknowledging this also helps to focus our thinking on what governments can do and how governments need to think in response to globalisation.

If the preceding economic geography arguments are correct, then New Zealand finds itself in a very difficult position (Guillemette, 2009), being quite literally on the edge of the global economy (Poot, 2004). These arguments imply that a fundamental re-thinking of New Zealand's role, position and potential in the twenty-first century global economy is required. Here, the first issue to be considered is the relationship between New Zealand and its major trading and labour market partner, Australia. Although Australia and New Zealand do not have a single internal market in the sense that the EU does, in many ways the institutional, cultural, historical and linguistic similarities between the two countries mean that the Australian and New Zealand markets are at least as integrated as those between many European countries. This is most obvious in terms of the almost complete Australian dominance of the New Zealand banking system, much of the retail sector, and also the free labour mobility between the two countries. The Closer Economic Relationship (CER) was originally established in 1983, and since then the economies of the two countries have become progressively integrated. As we have already seen, however, the labour productivity gap between the two countries initially only started

to emerge around 1985 and accelerated during the second half of the 1980s and the early 1990s (Maddison, 2006) and has not narrowed since then. The economic geography explanations of these (previously paradoxical) phenomena also imply that New Zealand increasingly exhibits economic characteristics that are typical of those of a region of Australia, or more properly, Australasia.

The problem with this scenario, that New Zealand increasingly exhibits economic characteristics that are typical of those of a region of Australasia, however, is that almost all of the positive externalities associated with the economic geography processes accrue to Australia itself. The most obvious case here is the fact that many of the productivity and human capital benefits of New Zealand's higher educational investment increasingly accrue to Australia, via out-migration. Yet, there is currently no compensating taxation and redistribution feedback mechanism from Australia to New Zealand, as would be the case for a region of a country (Poot, 1993).

Thinking of New Zealand increasingly as a region of Australasia therefore immediately points to three possible policy options: federation with Australia, a single currency between New Zealand and Australia, and also a single capital market.

The first option, federation with Australia, is clearly a political issue, which I will not discuss here, yet as is clear from the above arguments, it is an issue that will not go away in the long term.¹⁶

With regard to the second and third options, namely a single currency and capital market, while they will seem unrealistic and unpalatable to much of the New Zealand public, politicians and media, neither of these scenarios should be beyond debate, given the strength of the economic geography forces acting upon New Zealand. These issues have already been raised elsewhere (Grimes, 2005a,b), but for the purposes of this paper I will also steer clear of these issues as well in that they also involve complex monetary, financial, and optimal currency area issues. It is sufficient simply to note here that if changes such as a single currency or a single capital market significantly lower the transactions costs of investment flows from Australia or the rest of the world to New Zealand, then they are consistent with the policy logic derived from the economic geography arguments.

More generally, given that Australia sustains much larger agglomeration economies and a much larger home-market effect than New Zealand, the economic geography arguments suggest that in the current era of globalisation, the only other ways that New Zealand can possibly narrow the labour productivity and wage gaps between the two countries are by (A) increasing New Zealand's domestic agglomeration effects, and also (B) by trying radically to reduce all other spatial transactions costs between the two countries. The logic of the former approach is in terms of increasing local knowledge spillovers and innovations, while the logic of the latter approach is about maximising the scale of the backward knowledge spillovers and feedback effects from Australia to New Zealand. These two approaches are complementary and should be pursued simultaneously. Yet, each of these two economic approaches also implies radical policy agendas, some of which are very different to New Zealand's cultural perceptions of itself, many of which are very different from domestic aspirations regarding the ideal 'kiwi lifestyle', and all of which will be politically very difficult.

- (A1) In the case of promoting domestic agglomeration effects, the agenda here should focus primarily on increasing the scale of the Auckland-Hamilton-Tauranga triangular city-region (Le Heron & McDermott, 2006). By some

estimates Auckland is expected to grow by almost one third by 2021 (AREDS, 2008), although it may need to grow by up to 50% in order to make up the productivity gap with Australia's major urban areas. While Auckland's strategic infrastructure planning has been blighted by the high degree of governance fragmentation (METRO, 2006), infrastructure issues are critical for New Zealand (Grimes, 2008), and some progress has been made in terms of thinking about urban scale effect with regard to Auckland governance issues by Royal Commission proposals for the governance of Auckland. However, allowing Auckland to continue to grow implies that more land should be released for development in the Auckland region (McDermott, 2006) and also the public transport infrastructure (Allen Consulting Group, 2004; BERL, 2004) and services (Harward, 2009; Mees & Dodson, 2006) need to be significantly upgraded, in order to cope with the urban expansion. The growth of Auckland is also tied up with immigration policy. One of the easiest ways of stimulating Auckland's growth is via increased immigration, since as much as 70% of new immigrants choose to remain in Auckland (Sinclair, 2006). Many of these immigrants will also provide the possibility of developing and enhancing New Zealand's trade networks with their origin countries (Bryant et al., 2004). Yet, although increased immigration may not be a popular policy domestically, even Australia is considering adopting such a policy with increased urgency (Sheehan, 2008). This issue is further complicated by the fact that the divergence in labor productivity between the two countries actually makes Australia an increasingly more attractive option for migrants from third countries, and this is particularly so for highly-skilled workers.

- (A2) Arguments that consider Auckland to be a potential driver of New Zealand's exports (Skilling, 2006) are as yet far too premature. Auckland is currently not a major export base of its own accord, with an export-to-GDP ratio that is actually below that of other New Zealand regions (ARC, 2006), in this may well be the product of previous protection regimes (Gibson, 1993). Today, Auckland acts as primarily a knowledge conduit via which exports and imports of all New Zealand's sectors produced in all regions can be facilitated. As such, in commercial terms, Auckland is the dominant location where New Zealand connects with the rest of the global marketplace. Therefore, the primary focus here should be on upgrading the connectivity of Auckland to the rest of the country, as discussed in point (B1) below. Similarly, the role of the tertiary education system in promoting economic growth should be considered at a national level and not at the local Auckland urban level. The reason is that the most important mechanism by which universities contribute to regional knowledge flows is via the subsequent employment migration behaviour of their graduates around the country (Faggian & McCann, 2006, 2009; Simonen & McCann, 2008) rather than by direct local knowledge spillovers. As such, all universities in New Zealand play a critical role in fostering New Zealand's human capital and knowledge creation processes, and education policy should be focused on encouraging both specialisation and economies of scale across the whole tertiary education and crown research sector rather than focusing particularly on the University of Auckland. Achieving this combination in a country the size of New Zealand also implies that some level of rationalisation will be required

regarding the variety of activities undertaken by these research institutes, as New Zealand cannot maintain a broadly-based research agenda.

- (A3) The extreme export specialisation of New Zealand (World Bank, 2008b) means that the country's exports will still continue to be dominated by agriculture, tourism and educational services for decades to come. While New Zealand's terms of trade have remained fairly stable over the last two decades (Borkin, 2006), the issue here, therefore, is to find ways by which New Zealand can significantly upgrade the value-added levels in agricultural and tourism exports, industries with low productivity levels relative to many other sectors (Mason & Osborne, 2007). In spite of its high public profile, at present, labour productivity in New Zealand's agriculture is only ranked 16th in the world, and below that of the average level for the other rich countries (World Bank, 2009a). In terms of Fonterra and New Zealand's various other agricultural export organisations, a cooperative organisational structure will almost certainly limit the ability for innovation in these sectors. At the same time, although New Zealand's research and development (R&D) community is relatively specialised in agri-biosciences by international standards (OECD, 2006b), the absolute number of scientists involved in these activities in New Zealand is still tiny by global standards (OECD, 2006c). Yet, the importance of this work to the long-run prospects of New Zealand's growth cannot be overstated. While some commentators (Callaghan, 2009) argue that New Zealand's research funding should be blind with regard to the technology or sector, economic geography tells us that such a sector-neutral approach is only appropriate in a large and diversified market. Given New Zealand's particular structure and geography, promoting R&D and value-added in the agri-biosciences sector is critical (Oram, 2007). Once again, the challenge here relates to the entire tertiary education and crown research sector, and not just to one or two institutions. Finally, in terms of accessing overseas export markets, product branding, which emphasises the true environmental costs of New Zealand's products (Saunders et al., 2006), would also appear to be very important. Ironically, an increased concentration of population in Auckland and the upgrading of the city local public transport infrastructure ought to help improve New Zealand's overall carbon footprint, as urban density increases the efficiency of public transport usage.

In terms of reducing the spatial transactions costs between New Zealand and, most importantly, Australia, as well as the rest of the world, there are several potential policy approaches to be pursued here.

- (B1) Reasonable trans-Tasman competition for airlines exists (Hazledine, 2006). However, domestic air fares within New Zealand are extraordinary by international standards. Airfares between Auckland and Dunedin are often comparable to transatlantic airfares, while even Auckland to Wellington airfares are often equivalent to trans-European airfares. The standard argument as to why this is so is that this is because, in population terms, New Zealand has long and thin routes. Yet, long and thin routes are exactly the domain of low-costs carriers competing in Europe's contestable markets, specialising in routes such as Lodz to Limerick or La Rochelle to Southampton. In contrast, New Zealand's internal airline system is

dominated by a private company that simultaneously exhibits both monopoly and monopsony characteristics. This can never be even approximately efficient. Therefore, from the welfare perspective of the New Zealand economy as a whole, the domestic role and structure of Air New Zealand needs to be radically re-thought. The reason for this is that New Zealand's existing domestic air-transport structure unnecessarily penalises the geographically peripheral regions of New Zealand, by artificially increasing the travel costs between these regions and Auckland, New Zealand's major knowledge centre and gateway to the global economy. Given points (ii) and (iii) above, such regions will therefore be less able to play their potential role in today's globalisation processes than otherwise would be the case because their accessibility levels are artificially reduced. An observation such as this is difficult for many New Zealanders, because Air New Zealand is seen as a cultural icon, an international flag carrier that raises the country's profile overseas. Yet, the question still remains at what price are these cultural symbols being maintained? The abolition of the New Zealand Shipping Line monopoly the 1970s brought obvious advantages for New Zealand's exporters, and a similar logic needs to be applied to the domestic role of Air New Zealand. The role of Air New Zealand domestically and internationally can be seen as different things, and maintaining an international airline presence should not be dependent on maintaining internal domestic airfares of this order of magnitude. Breaking up the domestic air-transport monopoly is one strategy, or alternatively, given the inherent monopoly tendency of small economies (Sutton, 1991), greater regulation might be another strategy. From the perspective of the economy as a whole, however, the present structure is not tenable in the long-run.

Similar arguments also hold for Auckland Airport, which is also effectively a monopoly in New Zealand, and which according to some commentators achieves amongst the highest rates of return for any airport in the world (Eagles, 2006). As such, although the matter is now history, the previously-mooted takeover of Auckland International Airport by an Emirates-based airport management company may have offered New Zealand as a whole some new possibilities for increasing its international connectivity. In the case of air-transport markets in general, the experience of Europe has been that the dominant monopoly positions of both particular carriers and also particular airports has been rapidly reduced by the presence of low-cost contestable airlines choosing to use a variety of different airports and routes.

The fundamental point remains, however, that all of New Zealand's regions play a critical role in the country's exports, so anything that limits the accessibility, and therefore the potential global engagement of these regions, damages the economy as a whole. In particular, the role that New Zealand's monopolies play in increasing spatial transactions costs of accessing New Zealand's regions need to be reconsidered. Small markets are notorious for the development of monopolies (Sutton, 1991), and in the current era of globalisation, countries such as New Zealand must ensure that existing monopoly positions in transport and communications markets do not undermine overall competitiveness of the economy. This leads us naturally to the issue of telecommunications.

- (B2) The economic geography arguments above imply that a rapid upgrading of New Zealand's broadband infrastructure would appear to be crucial for reducing the spatial transactions costs of both international and also domestic interregional information flows (OECD, 2007b). With hindsight, however, the role and performance of New Zealand Telecom, and the failure of New Zealand to completely deregulate the telecommunications sector at a much earlier date, will probably prove to have been a national catastrophe for New Zealand (Newman, 2008). The reason is that at exactly the same time that globalisation processes were accelerating, and telecommunications markets were liberalising, New Zealand was largely standing still (Gerardin & Kerr, 2003). The arguments of Fujita et al. (1999) suggest that the small scale and isolation of New Zealand means that the spatial transactions costs of large-scale information flows should have been rapidly made even lower and at an earlier date in New Zealand than in other larger more centrally-located countries. On the other hand, instead of being in the vanguard of information and communications technologies, as the country was in the early 1990s (Newman, 2008), New Zealand is now in the bottom echelons of the communications technology rankings for advanced economies (Dutta & Mia, 2007), and Telecom's performance is almost certain to have been instrumental in this poor state of affairs (Newman, 2008).
- (B3) As a country, New Zealand is very open to global inward investment opportunities, such that New Zealand has the eighth highest levels of inward FDI relative to GDP in the OECD (OECD, 2008a). In contrast, in terms of the relative scale of its outward FDI, New Zealand is ranked only 22nd in the OECD (OECD, 2008a). Moreover, for comparison purposes, it is also ranked lower than any of the other small advanced OECD economies (OECD, 2008a). Meanwhile, with regard to its trade, the growth of New Zealand's trade relative to its GDP, 1993–2006, was the lowest for any OECD country (OECD, 2008a). Given its small size, the very low relative level of outward investment and the low overall growth in trade suggest that the New Zealand economy's active global engagement in overseas commerce really boils down to only a sub-section of New Zealand's firms and organisations. In part, this is probably because a very high proportion of New Zealand's firms are actually micro-firms by international standards, and are therefore simply too small to manage and maintain effective overseas control of commercial activities. Yet, adjusting for the scale and structure of the economy suggests that New Zealand is not particularly different to other OECD countries except to the extent that New Zealand's large companies are small by OECD standards (Mills & Timmins, 2004). Finding ways to increase the size of many New Zealand larger firms and organizations (Skilling & Boven, 2006a,b) so as to encourage outward global engagement (Williamson et al., 2006) may therefore help to promote outward investment. In addition, finding systems that also encourage New Zealand firms to move from simply exporting to overseas ownership, thereby promoting outward FDI and increasing New Zealand's global engagement, would appear to be critical. Such an approach can also be allied with an approach that targets particular types of inward investors. While 'picking winners' is widely regarded as having a poor history in much of New Zealand's industrial policy, using inward FDI-promotion strategies (Boston Consulting Group, 2001) to help to attract technologies regarded as

critical for New Zealand (and agri-biotechnology technologies would appear to be high on the list) is a pragmatic approach that is freely adopted by almost all of New Zealand's competitor countries.

Other potential policy issues (Guillemette, 2009), including those relating to trade and export promotion (McCann, 2003), and in particular with China and emerging Asian economies, have been discussed elsewhere (Stevens, 2007), so I will not discuss these further.

One final point relates to the desirability of using productivity as a benchmark in comparison with other measures of wellbeing and quality of life. These are complex issues that I will not discuss here, except with regard to one aspect which is directly related to the arguments contained in this paper. A possible counter argument to the thesis here is that the quality of life in New Zealand is higher than in other more crowded and urbanized countries because of the natural amenities available in New Zealand, and that the wellbeing or utility associated with a broad index of quality of life is not properly reflected in the productivity statistics. The logic of this argument is based on the observation that larger urban populations are always associated with greater population density, smaller dwelling units, and also longer average commuting times. As such, if increased agglomeration is associated with increased productivity, it may be that there is therefore something of a trade-off between the increased consumption associated with enhanced productivity and a reduced quality of life associated with density and congestion. Allowing for these broader quality of life arguments would narrow or even completely close the apparent gap between New Zealand and other countries.

There are, however, three responses to this argument, each of which relates to the economic geography arguments presented above. The first point is that New Zealand itself is not a low-urbanisation country by OECD standards (OECD, 2007d), with a relatively high proportion of its population living in urban areas, and with its major city facing very severe problems of traffic congestion compared with many cities in other countries (Harward, 2009). The second response is that natural amenities are largely unchanging over time, such that the combination of natural amenity consumption plus the consumption afforded by urban wages will still have fallen in New Zealand over time, relative to earlier decades.

The third response is that increasing urbanisation may itself also generate amenity benefits, a fundamental point overlooked by the above amenity argument. Cities have traditionally been viewed as having advantages in production but disadvantages in consumption (Glaeser et al., 2001). Yet, urban scale and density also allows for consumption opportunities that are not possible in other locations (Glaeser et al., 2001). While the availability of many consumption goods such as television and beer is largely independent of location, the availability of certain consumption possibilities such as high quality restaurants, theatre and boutique shops, do vary with location. We can regard these location-specific consumption possibilities as a component of man-made amenities. In addition, certain social amenities, such as societies and clubs, which are related to cultural and social networks, are also location-specific. From the perspective of an individual person, total utility in consumption is a function of the bundle of both the natural amenities and the man-made and social amenities (museums, libraries, cafes, restaurants, art galleries, orchestras, operas, societies and clubs) available at a particular location, as well as the range of consumption goods available at that

location. The fact that urban rents have increased faster than urban wages in many countries suggests that much of the current in-migration into cities in advanced economies is also related to an increasing consumer preference for urban amenity goods (Glaeser & Gottlieb, 2006; Glaeser et al., 2001) relative to the natural amenity goods available in non-urban areas. Both natural and man-made amenity goods across a variety of domestic and international locations can easily be consumed during vacations, and the feasible bundle of amenity goods accessible depends on the relationship between costs of transportation (including air travel) and the urban wages earned. As such, in order for the above natural-amenity argument to undermine the broad argument contained in this paper it would be necessary for us to argue that, over recent decades, the set of amenities available to New Zealand residents has increased relative to those available to residents of other countries by more than the productivity gap. There is no evidence of this. In fact, the arguments in this paper suggest exactly the opposite.

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Notes

1. See <http://siteresources.worldbank.org/DATASTATISTICS/Resources/GNIPC.pdf>
2. The Fraser Institute (2006) ranks New Zealand as the third freest economy in the world (<http://www.freetheworld.com/2006/EFWinternational-rls.pdf>). The World Heritage Foundation (2009) ranks NZ as the fifth freest economy in the world (<http://www.heritage.org/index/topten.cfm>).
3. www.transparency.org/layout/set/print/news_room/in_focus/2008/cpi2008/cpi_2008_table
4. Forbes ranked NZ tenth out of 135 countries in 2006 in terms of capital 'hospitality' (<http://www.forbes.com/lists/2006/6/CHI010.html>) and (http://www.forbes.com/2006/01/24/capital-hospitality-index_06caphosp_land.html).
5. We exclude here the micro-nations such as Andorra and San Marino, as well as tax haven states such as the Cayman Islands and Liechtenstein, or small oil states such as UAE, Kuwait or Qatar. On the other hand, we do here include Singapore and Hong Kong as comparators as they are both larger economies than New Zealand. Note that in terms of economic geography theory, however, both Singapore and Hong Kong are regarded as city-states whose labour productivity performance is higher than for many other countries, but which is actually lower than for many other large OECD cities (OECD, 2006a).
6. The Maddison (2006) estimates place Ireland ahead of NZ in 1995 whereas the OECD (2008a) estimates place NZ ahead of Ireland in 1995, and this accounts for the discrepancy in New Zealand's ranking of one place between the two sources. However, the trend pattern of New Zealand's relative ranking is not affected. The Penn World Table 6.2 places NZ above both the UK and Italy in 1985 whereas Maddison (2006) places both the UK and Italy above NZ. Hence, the 1985 NZ rank is 15 in the Penn World Table and 17 for Maddison (2006). Also, the 2004 OECD ranking places Spain above NZ, whereas the Penn World Table places NZ above Spain. This accounts for the discrepancy whereby in 2004 the OECD places NZ as the 23rd in the OECD (adjusted for PPP) whereas the Penn World Table 6.2 places New Zealand as 22nd.
7. The 2005 OECD report on New Zealand noted, the 'New Zealand economy has continued to expand at a vigorous clip ... bringing the average rate of growth since the recession of the early nineties to $3\frac{3}{4}$ per cent. More importantly, increases in real GDP per person have

- outpaced the OECD 10-year moving average since 2000 ...'. As such, the 'economy has continued on its strong upward course and living standards – measured as GDP per person – have risen steadily over the past decade, putting the country on track towards the government's objective of returning to the top half of the OECD' (OECD, 2005).
8. See <http://siteresources.worldbank.org/DATASTATISTICS/Resources/GNIPC.pdf>
 9. Yet, even this aspiration is slightly misleading in that the OECD grouping also includes the former transition economies of central Europe plus Mexico and Turkey, none of which are realistic comparison cases against which NZ should be benchmarking its performance, while it does not include Israel, Taiwan, Singapore or Hong Kong.
 10. Assuming that New Zealand is not overtaken in the meantime by countries that are already rapidly catching it up, such as South Korea or the Czech Republic, not to mention non-OECD countries such as Israel, Slovenia and Taiwan.
 11. NZ is ranked ninth in the world by the World Economic Forum (2007) for the overall condition of all of its institutions.
 12. Note that this argument is entirely consistent with the fact that transport costs have increased over recent years in comparison to the costs of manufactured goods, as the real costs of manufactured goods have fallen by more than the transport costs (OECD, 2008b). As such, in both marginal and elasticity terms, the effects of transport costs have actually become more significant over recent years.
 13. In absolute terms, New Zealand is the third largest overseas residence destination for Australian citizens after the UK and USA, with half as many Australian residents as UK and three quarters of the total US number. Relative to its size, New Zealand is therefore by far the biggest overseas residence destination for Australian citizens (Dumont & Lemaitre, 2005).
 14. <http://www.population.govt.nz/myth-busters/MythDiaspora.htm>.
 15. <http://www.stats.govt.nz/census/2006-census-data/regional-summary-tables.htm>.
 16. Moreover, it is a possibility that is still allowed for in the Australian Constitution.

References

- Acs, Z.J. (2002). *Innovation and the growth of cities*. Cheltenham: Edward Elgar.
- Acs, Z.J., & Szerb, L. (2009). The Global Entrepreneurship Index (GEINDEX) and the stages of economic development. *Global Entrepreneurship Monitor*, <http://www.gemconsortium.org>
- Acs, Z.J., Bosma, N., & Sterberg, R. (2008). *The entrepreneurial advantage of world cities: Evidence from global entrepreneurship monitor data*. Working Paper, H200810, SCALES, University of Utrecht and Netherlands Ministry of Economic Affairs.
- Alesina, A., & Spolaore, E. (2005). *The size of nations*. Cambridge MA: MIT Press.
- Allen Consulting Group (2004). *Benefits of investing in New Zealand's road infrastructure*. Sydney: Allen Consulting Group.
- AREDS (2008). *Growing Auckland: Auckland regional economic development strategy*. Auckland: Committee for Auckland.
- ARC (2006). *Metro Project Action Plan: Implementing the Auckland regional development strategy*. Auckland: Auckland Regional Council.
- Auckland City Council (2006). *The economy of Auckland's CBD 2005/2006*. Auckland: Auckland City Council.
- Barro, R., & Sala-i-Martin, X. (1992). Convergence. *Journal of Political Economy*, 100, 223–251.
- Battersby, B. (2006). *The effect of geographic isolation on productivity levels*. Treasury Working Paper 2006-03, Canberra.
- Baxendine, S., Cochrane, W., & Poot, J. (2006). *Description and spatial analysis of employment change in New Zealand regions 1986–2001*. Population Studies Centre Discussion Papers No. 57, University of Waikato.
- BERL (2004). *Investing for growth: Economic and strategic importance of the eastern transport corridor*. BERL Report 4297, Wellington.
- Berry, C., & Glaeser, E.L. (2005). The divergence of human capital levels across cities. *Papers in Regional Science*, 84(3), 407–444.
- Birrel, B., Rapson, V., & Smith, T.F. (2006). *Australia's net gains from international skilled movement*. Centre for Population and Urban Research, Monash University, Melbourne.

- BITRE (2008). *A regional economy: A case study of Tasmania*. Australian Government Bureau of Infrastructure, Transport and Regions, Report 116, Department of Infrastructure, Transport, Regional Development and Local Government, Canberra.
- BITRE (2009). *Household wealth*. Information Paper 63. Australian Government Bureau of Infrastructure, Transport and Regions, Report 116, Department of Infrastructure, Transport, Regional Development and Local Government, Canberra.
- Borkin, P. (2006). *Past, present and future developments in New Zealand's terms of trade*. New Zealand Treasury Working Paper 06-09, Wellington.
- Borts, G.H., & Stein, J.L. (1964). *Economic growth in a free market*. New York: Columbia University Press.
- Boston Consulting Group (2001). *Building the future: Using FDI to help fuel New Zealand's prosperity*. Auckland: Boston Consulting Group.
- Bradley, R., & Gans, J.S. (1998). Growth in Australian cities. *Economic Record*, 74, 266–278.
- Brakman, S., & van Marrewijk, C. (2008). It's a big world after all: on the economic impact of location and distance. *Cambridge Journals of Regions, Economy and Society*, 1(3), 411–437.
- Breschi, S. & Malerba, F. (Eds.). (2005). *Clusters, networks, and innovation*. Oxford: Oxford University Press.
- Bryant, J., & Law, D. (2004). *New Zealand's diaspora and overseas-born population*. New Zealand Treasury Working Paper 04/13, New Zealand Treasury, Wellington.
- Bryant, J., Genc, M., & Law, D. (2004). *Trade and migration to New Zealand*. New Zealand Treasury Working Paper 04/18, New Zealand Treasury, Wellington.
- BTRE (2004). *Focus on Regions: Taxable Income*. BTRE Information Paper 54, Department of Transport and Regional Services. Canberra, Australia.
- Callaghan, P. (2009). *Wool to Weta: Transforming New Zealand's culture and economy*. Auckland: Auckland University Press.
- Carlino, G.A., Chatterjee, S., & Hunt, R.M. (2007). Urban density and the rate of invention. *Journal of Urban Economics*, 61(3), 389–419.
- Ciccone, A., & Hall, R.E. (1996). Productivity and the density of economic activity. *American Economic Review*, 86, 54–70.
- Clark, G.L., Feldmann, M.P., & Gertler, M.S. (2000). *The Oxford handbook of economic geography*. Oxford: Oxford University Press.
- Collier, P. (2007). *The bottom billion: Why the poorest countries are failing and what can be done about it*. Oxford: Oxford University Press.
- Combes, P.P. (2000). Economic structure and local growth: France, 1984–1993. *Journal of Urban Economics*, 47, 329–355.
- Combes, P-P., & Lafourcade, M. (2005). Transport costs: measures, determinants, and regional policy implications for France. *Journal of Economic Geography*, 319–349.
- Conway, P., & Orr, A. (2000). The process of economic growth in New Zealand. *Reserve Bank of New Zealand Bulletin*, 63(1), 4–20.
- Corporation of London (CoL) (2008). *The Global Financial Centres Index 3*. London: Corporation of London.
- Crafts, N. (2004). Globalization and economic growth: a historical perspective. *The World Economy*, 27(1), 45–58.
- Crawford, R. (2004). *The effects of agglomeration on economic activity: The empirical evidence on mechanisms and magnitudes*. Working Paper, Ministry of Economic Development, Wellington.
- Crawford, R., Fabling, R., Grimes, A., & Bonner, N. (2007). National R&D and patenting: is New Zealand an outlier? *New Zealand Economic Papers*, 41(1).
- Crocombe, G.T., Enright, M.J., & Porter, M.E. (1991). *Upgrading New Zealand's competitive advantage*. Auckland: Oxford University Press.
- Dalziel, P. (2002). New Zealand's economic reforms: an assessment. *Review of Political Economy*, 14(1), 31–46.
- Davis, G., & Ewing, P. (2005). *Why has Australia done better than New Zealand? Good luck or good management?* Australian Treasury Working Paper. Canberra.
- Davis, G., & Rahman, J. (2006). *Perspectives on Australia's productivity prospects*. Treasury Working Paper, 2006-04, Canberra.

- Disdier, A.-C., & Head, K. (2007). The puzzling persistence of the distance effect on bilateral trade. *Review of Economics and Statistics*, 90(1), 37–48.
- Dorling, D., & Thomas, B. (2004). *People and places: A 2001 census atlas of the UK*. Bristol: Policy Press.
- Dumont, J.-C., & Lemaitre, G. (2005). *Counting immigrants and expatriates in OECD countries: A new perspective*. Directorate for Employment Labour and Social Affairs DELSA. Paris: OECD.
- Duranton, G., & Puga, D. (2001). Nursery cities: urban diversity, process innovation, and the life cycle of products. *American Economic Review*, 91(5), 1454–1477.
- Duranton, G., & Storper, M. (2007). Rising trade costs? Agglomeration and trade with endogenous transaction costs. *Canadian Journal of Economics*, 41(1), 292–319.
- Dutta, S., & Mia, I. (2007). *World economic forum: The global information technology report 2006–2007: Connecting the networked economy*. London: Palgrave.
- Eagles, J. (2006). Can a monopoly company be trusted. *The New Zealand Herald*, 31 July 2006.
- Easton, B. (1994). Economic and other ideas behind the New Zealand reforms. *Oxford Review of Economic Policy*, 10(3), 78–94.
- Easton, B. (2007). *Globalisation and the Wealth of Nations*. Auckland: Auckland University Press.
- Eichengreen, B. (2007). *The European economy since 1945: Coordinated capitalism and beyond*. Princeton: Princeton University Press.
- Evans, L., Grimes, A., Teece, D., & Wilkinson, B. (1996). Economic reform in New Zealand 1984–95: The pursuit of efficiency. *Journal of Economic Literature*, 34, 1856–1902.
- Faggian, A., & McCann, P. (2006). Human capital flows and regional knowledge assets: A simultaneous equation approach. *Oxford Economic Papers*, 58(3), 475–500.
- Faggian, A., & McCann, P. (2009). Human capital, graduate migration and innovation in British regions, 2008. *Cambridge Journal of Economics*, forthcoming.
- Faggian, A., McCann, P., & Sheppard, S.C. (2007). Some evidence that women are more mobile than men: Gender differences in UK graduate migration behaviour. *Journal of Regional Science*, 47(3), 517–539.
- Feldman, M.P., & Massard, N. (2001). *Institutions and systems in the geography of innovation*. Amsterdam: Kluwer.
- Findlay, R., & O'Rourke, K.H. (2007). *Power and plenty: Trade, war, and the world economy in the second millennium*. Princeton: Princeton University Press.
- Fox, K.J. (2005). *Returns to scale, technical progress and total factor productivity growth in New Zealand industries*. New Zealand Treasury Working Paper, 05/04, Wellington.
- Fraser Institute (2006). *Economic freedom promotes growth while foreign aid fails developing countries*. Downloaded from <http://www.freetheworld.com/2006/EfWinternational-rls.pdf>
- Friedman, T.L. (2007). *The world is flat: A brief history of the twenty-first century* (3rd edn). New York: Picador.
- Fujita, M. (Ed.). (2007a). *Regional integration in East Asia from the viewpoint of spatial economics*. Basingstoke, UK: Palgrave.
- Fujita, M. (2007b). Globalization, regional integration, and spatial economics: An introduction. In M. Fujita (Ed.), *Regional integration in East Asia from the viewpoint of spatial economics*. Basingstoke, UK: Palgrave.
- Fujita, M. (2007c). Development of East Asian regional economies: A view from spatial economics. In M. Fujita (Ed.), *Regional integration in East Asia from the viewpoint of spatial economics*. Basingstoke, UK: Palgrave.
- Fujita, M., Krugman, P., & Venables, A.J. (1999). *The spatial economy: Cities regions and international trade*. Cambridge, MA: MIT Press.
- Fujita, M., & Thisse, J.-F. (2002). *Economics of agglomeration: Cities, industrial location, and regional growth*. Cambridge: Cambridge University Press.
- Gaspar, J., & Glaeser, E.L. (1998). Information technology and the future of cities. *Journal of Urban Economics*, 43, 136–156.
- Gerardin, D., & Kerr, M. (2003). *Controlling market power in telecommunications*. Oxford: Oxford University Press.
- Gibson, J. (1993). Regional consequences of New Zealand industry assistance policies. *New Zealand Journal of Geography*, April, 2–6.

- Glaeser, E.L. (1997). Are cities dying? *Journal of Economic Perspectives*, 12(2), 139–160.
- Glaeser, E.L. (2008). *Cities, agglomeration and spatial equilibrium*. Oxford: Oxford University Press.
- Glaeser, E.L., & Gottlieb, J.D. (2006). Urban renaissance and the consumer city. *Urban Studies*, 43(8), 1275–1299.
- Glaeser, E.L., & Kohlhase, J. (2004). Cities, regions and the decline of transport costs. *Papers in Regional Science*, 83(1), 197–228.
- Glaeser, E.L., Kolko, J., & Saiz, A. (2001). Consumer city. *Journal of Economic Geography*, 1(1), 27–50.
- Glaeser, E., Kallal, H., Scheinkman, J., & Shleifer, A. (1992). Growth in cities. *Journal of Political Economy*, 100, 1126–1152.
- Gould, B. (2008). *Rescuing the New Zealand economy*. Nelson: Craig Potton Publishing.
- Gould, J. (1982). *The rake's progress: The New Zealand economy since 1945*. Auckland: Hodder and Stoughton.
- Gravitas (2003). *Business location in relation to the CBD*. Report Prepared for Auckland City Council, Auckland.
- Grimes, A. (2005a). New Zealand: A typical Australasian economy? In A. Ladley & J. Martin (Eds.), *The visible hand: The changing role of the state in New Zealand*. Wellington: Institute of Policy Studies.
- Grimes, A. (2005b). Regional industry cycles in Australasia: Implications for a common currency. *Journal of Asian Economies*, 16(3), 380–397.
- Grimes, A. (2008). *The role of infrastructure in developing New Zealand's economy*. Motu Economic and Public Policy Research Paper, Wellington.
- Grimes, A., & Liang, Y. (2008). *Bridge to somewhere: The value of Auckland's northern motorway extensions*. Motu Working Paper 08-07, Motu Economic and Public Policy Research, Wellington.
- Guillemette, Y. (2009). *Structural policies to overcome geographic barriers and create prosperity in New Zealand*. OECD Economics Department Working Paper, No 696. Paris: Organisation for Economic Cooperation and Development.
- Gyngell, A., Skilling, D., & Thirlwell, M. (2007). *Australia and New Zealand in a globalising world*. Lowy institute perspectives. Sydney: Lowy Institute for International Policy.
- Hall, J., & Casey, A. (2006). *International comparative surveys of regulatory impact*. New Zealand Treasury Working Paper 06-05, Wellington.
- Hall, J., & Scobie, G. (2005). *Capital shallowness: A problem for New Zealand*. New Zealand Treasury Working Paper 05/05, Wellington.
- Harbridge, R. (1993). *Employment contracts: The New Zealand experience*. Wellington: Victoria University Press.
- Harrington, A. (2005). *The contribution of the primary sector to New Zealand's economic growth*. New Zealand Treasury Working Paper 05/04, Wellington.
- Harris, R.I.D., & Daldy, B.M. (1994). *Labor market adjustment in New Zealand*. Aldershot: Avebury.
- Harward, E. (2009). Bright lights, big mess. *Sunday Star Times*, 29 March 2009.
- Hazledine, T. (2006). *Pricing and competition in Australasian air travel markets*. Working Paper, Department of Economics, University of Auckland.
- Henderson, J.V., & Thisse, J.-F. (2004). *Handbook of regional and urban economics volume 4: Cities and geography*. Amsterdam: Elsevier North-Holland.
- Hendy, S., & Sissons, C. (2009). *Innovators, innovation and increasing returns to scale: What can we learn from patent databases*. Paper presented at MoRST Chatshop, 5 June 2009, Wellington.
- Iammarino, S., & McCann, P. (2006). The structure and evolution of industrial clusters: Transactions, technology and knowledge spillovers. *Research Policy*, 35, 1018–1036.
- IMF (2004). *New Zealand: Selected issues*. IMF Country Report No 04/127, International Monetary Fund, Washington DC.
- Karagedikli, O., Maré, D., & Poot, J. (2000). Disparities and despair: Changes in regional income distributions in New Zealand 1981–1996. *Australasian Journal of Regional Studies*, 6(3), 323–347.
- Kambhampati, U., & McCann, P. (2007). The regional performance and characteristics of Indian manufacturing industry. *Regional Studies*, 41(3), 281–294.

- Kasper, W. (2002). Losing sight of the lodestar of economic freedom: A report card on New Zealand's economic reforms. *New Zealand Business Roundtable*, December.
- Krugman, P. (1991). *Geography and trade*. Cambridge, MA: MIT Press.
- Krugman, P. (1997). *The age of diminished expectations*. Cambridge, MA: MIT Press.
- Krugman, P. (2007). The 'new' economic geography: Where are we? In M. Fujita (Ed.), *Regional integration in East Asia from the viewpoint of spatial economics*. Basingstoke, UK: Palgrave.
- Krugman, P., & Venables, A.J. (1995). Globalization and the inequality of nations. *Quarterly Journal of Economics*, 110(4), 857–880.
- Law, D., & McLellan, N. (2005). *The contributions from firm entry, exit and continuation to labour productivity growth in New Zealand*. New Zealand Treasury Working Paper 05-01, Wellington.
- Layard, R. (2005). *Happiness: Lessons from a new science*. London: Allen Lane.
- Le Heron, R., & McDermott, P. (2006). *A framework for Auckland's economic development*. Paper Prepared for the Metropolitan Auckland Project Team, Auckland.
- Lewis, G. (2002). *The economic transformation debate: Where have we got to?* Paper presented at the New Zealand Association of Economists Conference, 26–28 June, Wellington.
- Lewis, G., & Stillman, S. (2005). *Regional economic performance in New Zealand: How does Auckland compare?* New Zealand Treasury Working Paper 05/08, Wellington.
- Lewis, G., & Stillman, S. (2007). Regional economic performance in New Zealand: How does Auckland compare? *New Zealand Economic Papers*, 41(4).
- Maddison, A. (2006). *The world economy volume 1: A millennial perspective volume 2: Historical statistics*. Paris: OECD.
- Mason, G., & Osborne, M. (2007). *Productivity, capital-intensity and labour quality at sector level in New Zealand and the UK*. New Zealand Treasury Working Papers 07-01, Wellington.
- Maré, D.C. (2008). *Labor productivity in Auckland firms*. Motu Working Paper 08–12, Motu Economic and Public Policy Research, Wellington.
- Mastercard (2008). *Worldwide centers of commerce index*. www.mastercardworldwide.com.
- McCann, P. (2003). *Geography, trade and growth: Problems and possibilities for the New Zealand economy*. New Zealand Treasury Working Paper 03.03, Wellington.
- McCann, P. (2007). Sketching out a model of innovation, face-to-face interaction and economic geography. *Spatial Economic Analysis*, 2(2), 117–134.
- McCann, P. (2008). Globalization and economic geography: The world is curved, not flat. *Cambridge Journal of Regions, Economy and Society*, 1(3), 351–370.
- McCann, P. (2009). Globalisation, multinationals and the BRIICS countries. In R. Lattimore & R. Safadi (Eds.), *Globalisation and emerging economies*. Paris: OECD.
- McCann, P., & Acs, Z.J. (2009). *Globalisation: Countries, cities and multinationals*. Working Paper, Department of Economics, University of Waikato.
- McCann, P., & Arita, T. (2006). Clusters and regional development: Some cautionary observations from the semiconductor industry. *Information Economics and Policy*, 18(2), 157–180.
- McCann, P., Poot, J., & Sanderson, L. (2009). *Migration, relationship capital and international travel: Theory and evidence*. Working Paper, Department of Economics, University of Waikato.
- McDermott, P. (2006). *Review of business land capacity in the Auckland region*. Paper Prepared for the Metropolitan Auckland Project Team, Auckland.
- McDonald, P., & Temple, J. (2006). *Immigration and the supply of complex problem solvers in the Australian economy*. Australian Government Department of Immigration and Multicultural Affairs, Canberra.
- MED (2005). *Economic development indicators 2005: Growth through innovation: Sustainable economic growth for all New Zealanders*. Ministry of Economic Development and The Treasury, Wellington.
- MED (2007). *Economic development indicators*. Ministry of Economic Development, The Treasury and Statistics New Zealand, Wellington.
- Mees, P., & Dodson, J. (2006). *Backtracking Auckland: Bureaucratic rationality and public preferences in transport planning*. Urban Research Program Issues Paper No. 5, Griffith University, Brisbane.
- METRO (2006). *International review of Auckland metro region*. Paper Prepared for the Auckland Metro Project Team, June.

- Mills, D., & Timmins, J. (2004). *Firm dynamics in New Zealand: A comparative analysis with OECD countries*. New Zealand Treasury Working Paper 04-11, Wellington.
- Newman, K. (2008). *Connecting the clouds: The internet in New Zealand*. Auckland: Activity Press.
- Nijkamp, P., & Poot, J. (1998). Spatial perspectives on new theories of economic growth. *Annals of Regional Science*, 32(1), 7–38.
- NZIER (2006a). *The New Zealand–Australian income differential*. NZIER Working Paper 2006/05, Wellington.
- NZIER (2006b). *Auckland's contribution to the government's surplus in 2005: Estimating the net flow of central government funding from the Auckland region in fiscal year 2005*. Wellington: NZIER.
- OECD (2003). *Economic surveys: New Zealand*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2005). *OECD economic surveys New Zealand*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2006a). *Competitive cities in the global economy*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2006b). *OECD surveys: Australia*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2006c). *OECD biotechnology statistics 2006*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2007a). *OECD economic surveys: New Zealand*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2007b). *OECD reviews of innovation policy: New Zealand*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2007c). *OECD science, technology and industry scoreboard 2007*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2007d). *OECD regions at a glance*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2008a). *OECD factbook 2008: Economic, environmental and social statistics*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2008b). *Economic policy reforms: Going for growth*. Paris: Organisation for Economic Cooperation and Development.
- OECD (2008c). *The contribution of economic geography to GDP per capita*. Economics Department Working Paper. Paris: Organisation for Economic Cooperation and Development.
- Oram, R. (2007). *Reinventing paradise: How New Zealand is starting to earn a bigger, sustainable living in the world economy*. North Shore: Penguin Books.
- Polenske, K. (Ed.). (2007). *The economic geography of innovation*. Cambridge: Cambridge University Press.
- Poot, J. (1993). Trans-Tasman migration and economic growth in Australasia. In G. Carmichael (Ed.), *Trans-Tasman migration: Trends, causes and consequences*. Canberra: Australian Government Publishing Service.
- Poot, J. (Ed.). (2004). *On the edge of the global economy*. Cheltenham: Edward Elgar.
- Poot, J. (2008). *Drivers and economic impacts of Auckland demographic change*. Paper Presented at the Symposium on Auckland Economy, 8 October 2008. University of Waikato.
- Poot, J. (2009). Trans-Tasman migration, transnationalism and economic development in Australasia. *Asian and Pacific Migration Journal*, forthcoming.
- Porter, M.E. (1990). *The competitive advantage of nations*. New York: Free Press.
- Porter, M.E., Sala-i-Martin, X., & Schwab, K. (2007). *World economic forum: The global competitiveness report*. Basingstoke: Palgrave.
- Procter, R. (2008). *Inside the black box: Policies for economic growth*. Ministry of Economic Development Occasional Paper, Wellington.
- Rodrik, D. (2007). *One economics many recipes*. Princeton, NJ: Princeton University Press.
- Rowe, J.E. (2006). Is Auckland and entrepreneurial or global city? *Town Planning Review*, 77(5), 583–604.
- Rugman, A. (2005). *The regional multinationals: MNEs and 'global' strategic management*. Cambridge: Cambridge University Press.
- Saunders, C., Barber, A., & Taylor, G. (2006). *Food miles – comparative energy/emissions performance of New Zealand's agriculture industry*. Lincoln University: AERU.

- Shallcrass, R. (2006). *Family silver*. Wellington: Victoria University Press.
- Sheehan, P. (2008). And the challenge of migration? *Sydney Morning Herald*, 4 August 2008.
- Simonen, J., & McCann, P. (2008). Firm innovation: The influence of R&D cooperation and the geography of human capital inputs. *Journal of Urban Economics*, 64(1), 146–154.
- Sinclair, R. (2006). *Immigration policy analysis*. Metro Project, Auckland Regional Council, Auckland.
- Singleton, J. (2006). *Innovation and independence: The Reserve Bank of New Zealand 1973–2002*. Auckland: Auckland University Press.
- Silverstone, B., Bollard, A., & Lattimore, R. (1996). *A study of economic reform: The case of New Zealand*. Amsterdam: North-Holland.
- Simonen, J., & McCann, P. (2008). Firm innovation: The influence of R&D cooperation and the geography of human capital inputs. *Journal of Urban Economics*, 64(1), 146–154.
- Simmons, G. (2004). The impact of scale and remoteness on New Zealand's industrial structure and firm performance. In J. Poot (Ed.), *On the edge of the global economy*. Cheltenham: Edward Elgar.
- Skilling, D. (2001a). *The importance of being enormous: Towards an understanding of the New Zealand economy*. Economic Transformation Project, New Zealand Treasury, Wellington.
- Skilling, D. (2001b). *Comparative disadvantage: NZ's economic growth in historical perspective*. Economic Transformation Project, New Zealand Treasury, Wellington.
- Skilling, D. (2006). *The role of Auckland in meeting New Zealand's productivity challenge*. Essay: New Zealand Institute, Auckland.
- Skilling, D., & Boven, D. (2005a). *No country is an island: Moving the New Zealand economy by taking it to the world*. Discussion Paper 2005-03, New Zealand Institute, Auckland.
- Skilling, D., & Boven, D. (2005b). *Dancing with the stars: The international performance of the New Zealand economy*. Discussion Paper 2005-04, New Zealand Institute, Auckland.
- Skilling, D., & Boven, D. (2006a). *The flight of the Kiwi: Going global from the end of the world*. Discussion Paper 2006-01, New Zealand Institute, Auckland.
- Skilling, D., & Boven, D. (2006b). *Developing Kiwi global champions: Growing successful New Zealand multinational companies*. Discussion Paper 2006-03, New Zealand Institute, Auckland.
- Smart, W. (2006). *Do student loans drive people overseas – what is the evidence?* Wellington: Ministry of Education.
- Statistics New Zealand (2005). *New Zealand in the OECD*. Statistics New Zealand: Wellington.
- Stevens, W. (2007). *The risks and opportunities from globalisation*. New Zealand Treasury Working Paper 07-05, Wellington.
- Storper, M., & Venables, A.J. (2004). Buzz: Face-to-face contact and the urban economy. *Journal of Economic Geography*, 4, 351–370.
- Sutton, J. (1991). *Sunk costs and market structure*. Cambridge, MA: MIT Press.
- Szeto, K.L., & McLoughlin, S. (2008). *Does quality matter in labour input? The changing pattern of labour composition in New Zealand*. New Zealand Treasury Working Paper 08-01, Wellington.
- Venables, A.J. (2006). Shifts in economic geography and their causes. *Federal Reserve Bank of Kansas City Economic Review*, 91(4), 61–85.
- Williamson, J., de Blaauw, N., & Waite, D. (2006). *Metropolitan Auckland: Background paper*. Auckland: Ascari Partners.
- World Bank (2008a). *2008 world development indicators*. Washington, DC: World Bank.
- World Bank (2008b). *World trade indicators 2008: Benchmarking policy and performance*. Washington, DC: World Bank.
- World Bank (2009a). *Doing business 2009*. Washington, DC: World Bank.
- World Bank (2009b). *Reshaping economic geography: World development report*. Washington, DC: World Bank.
- World Economic Forum (2007). *Global competitiveness report 2007–2008*. New York: Palgrave Macmillan.
- World Heritage Foundation (2009). *2009 index of economic freedom: The link between economic opportunity and prosperity*. Washington, DC: Heritage Books.